Module 4.1

Total Survey Error Framework

Day 4: Advanced Error Mitigation & Quality Assurance

SADC Regional Training Workshop

5 March 2026

The Monday Morning Crisis

[Z] Zambara Case

Lindiwe arrives at the Zambara National Statistics Office to find her inbox flooded with urgent messages. The Field Monitoring Dashboard displays cascading **red alerts** across multiple provinces...

- **Crisis 1: Missing Income**
  - Item non-response: **24.4%**
  - Northern Cape worst: 31.6%
  - Pattern suggests *not random*
- **Crisis 2: Inaccessible PSUs**
  - 12 PSUs unreachable
  - Eastern Forests region
  - Civil unrest ongoing

Before fixing problems, Lindiwe needs a systematic framework...

Learning Objectives

Module 4.1 Objectives

By the end of this module, you will be able to:

1. **Define** Total Survey Error and its component sources
2. **Distinguish** sampling errors from non-sampling errors
3. **Apply** the TSE framework to diagnose survey quality issues
4. **Design** monitoring dashboards for error detection

Reference: Groves, R.M. et al. (2009). *Survey Methodology*. 2nd ed. Wiley.

What is Total Survey Error?

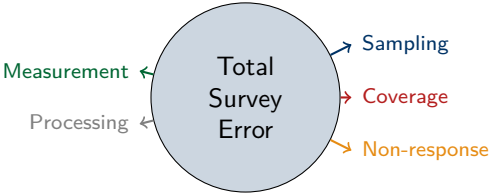
Definition

Total Survey Error (TSE) is the accumulation of all errors that may arise in the design, collection, processing, and analysis of survey data.

The TSE Principle:

- Quality $\neq$ sample size alone
- Multiple error sources compound
- Some errors visible, many hidden
- Framework enables systematic diagnosis

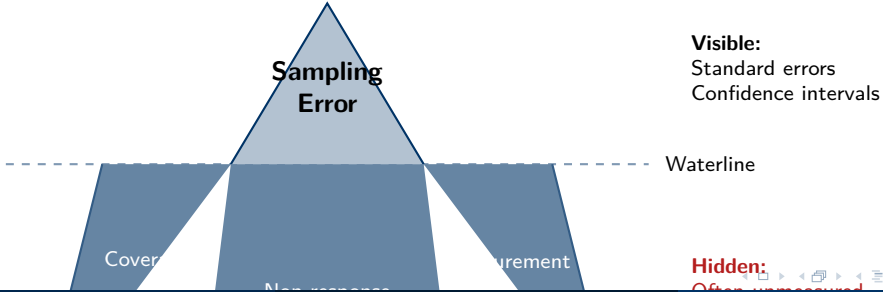
Source: Groves et al. (2009), Chapter 2



The TSE Iceberg

*** Key Insight**

Sampling error is the tip of the iceberg — visible and quantifiable.
Non-sampling errors lurk beneath the surface — often larger and hidden.



Visible:
Standard errors
Confidence intervals

Hidden:

TSE Decomposition: The Formula

Mean Squared Error Decomposition

$$MSE(\hat{\theta}) = \underbrace{Var(\hat{\theta})}_{\text{Variance}} + \underbrace{[E(\hat{\theta}) - \theta]^2}_{\text{Bias}^2}$$

Total Survey Error encompasses both:

$\sim$ **Variance**

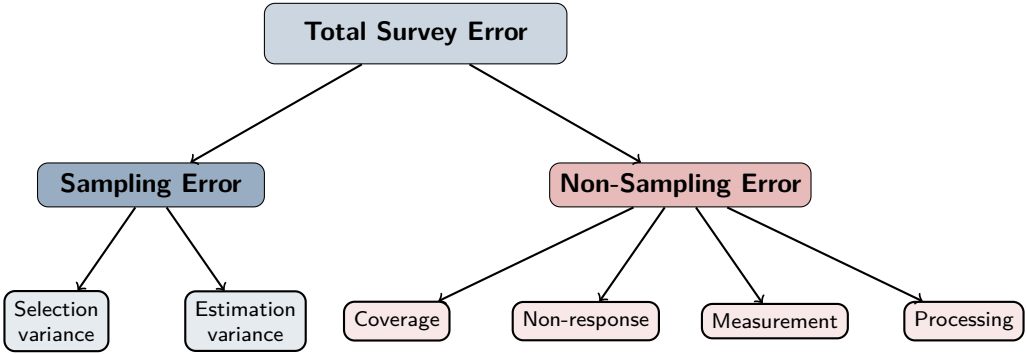
- Random fluctuation
- Decreases with sample size
- *Measurable* from data

$\rightarrow$ **Bias**

- Systematic deviation
- Does NOT decrease with n
- Often *unmeasurable*

! Warning

Two Families of Error



✓ **Quantifiable**
Standard errors, design effects

× **Often Hidden**
Requires auxiliary information

The Dual Nature of Error

Representation Errors

Who is in the sample?

- Coverage error
- Sampling error
- Non-response error

Does the sample represent the population?

Measurement Errors

What values are recorded?

- Validity error
- Reliability error
- Processing error

Are the measurements accurate?

* Key Insight

Lindiwe's crises span **both** dimensions:

Coverage Error

Definition

Coverage error arises when the sampling frame does not perfectly match the target population — through undercoverage, overcoverage, or duplication.

[-] Undercoverage

Population members missing from frame

Example: New settlements not in census

[+] Overcoverage

Non-members included in frame

Example: Demolished dwellings still listed

[=] Duplication

Units listed multiple times

Example: Same HH in two EAs

Reference: Lohr (2022), Section 8.5

Coverage Error in Zambara

[Z] Zambara Case

Lindiwe examines the sampling frame and discovers it's based on the 2020 census. Since then, three informal settlements have grown around the Mining Belt capital...

Coverage Bias Formula:

$$Bias_{cov} = \frac{N_{NC}}{N} \times (\bar{Y}_{NC} - \bar{Y}_C)$$

Where:

- N_{NC} Non-covered population size
- N Total population size
- $\bar{Y}_{NC}$ Mean of non-covered units

! Warning

If non-covered units differ systematically from covered units, estimates are biased!

Sampling Error

Definition

Sampling error is the difference between a sample estimate and the population parameter due to observing only a subset of the population.

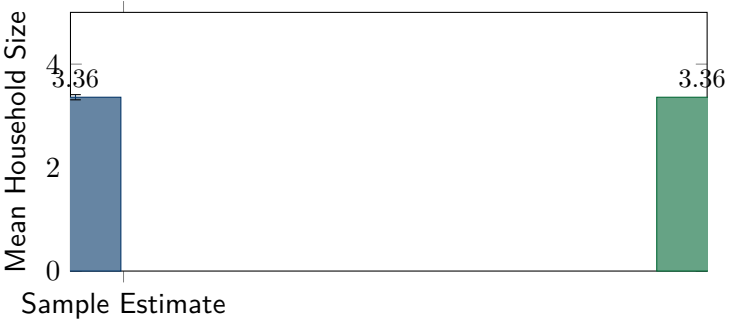
Key Properties:

- Inherent to *all* probability samples
- **Decreases** with sample size: $SE \propto 1/\sqrt{n}$
- **Quantifiable** through variance estimation
- Reported via standard errors and confidence intervals

*** Key Insight**

Sampling Error: What We See

Zambara HLCS: Estimated Mean Household Size



$\hat{\bar{Y}} = 3.36$ persons SE = 0.05 95% CI: [3.26, 3.46]

Sampling error alone suggests excellent precision — but what about bias?

Non-Response Error

Definition

Non-response error occurs when sampled units fail to provide data, and respondents differ systematically from non-respondents.

[U] Unit Non-Response

- Entire questionnaire missing
- Refusals, non-contacts
- Affects sample composition

[?] Item Non-Response

- Specific questions unanswered
- "Don't know", refusal
- Affects variable completeness

Non-Response Bias Formula:

$$Bias_{NR} = (1 - RR) \times (\bar{Y}_R - \bar{Y}_{NR})$$

Zambara Response Rate Dashboard

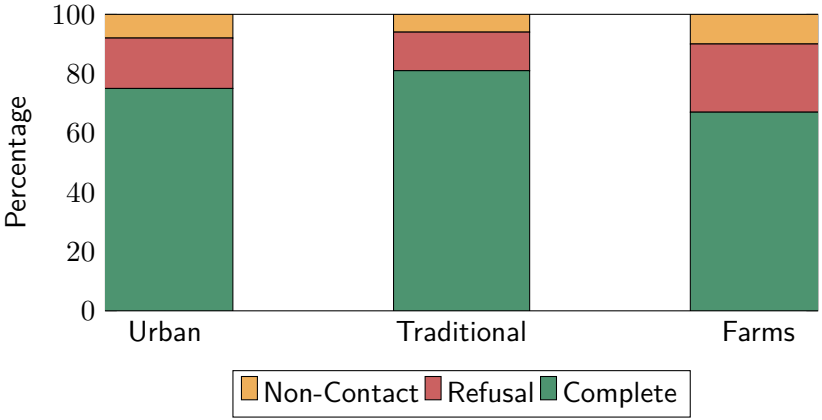
[Z] Zambara Case

Lindiwe pulls up the Field Monitoring System. The national response rate shows **75.6%** — below the 85% target. But the real story is in the details...

Province	Total	Complete	RR1%	Refusal%	NC%
Northern Cape	855	585	**68.4**	21.5	10.1
Eastern Cape	2,706	1,989	73.5	18.4	8.1
Gauteng	5,526	4,156	75.2	16.7	8.1
KwaZulu-Natal	3,257	2,516	77.2	15.3	7.5
Limpopo	2,385	1,870	**78.4**	14.8	6.8
NATIONAL	**20,940**	**15,839**	**75.6**	**16.5**	**7.9**

Source: Lab 4.1 Output (Simulated Data)

Response Patterns by Geography Type



[C] Urban: Higher refusals
Privacy concerns, gatekeeping

[F] Farms: Higher non-contact
Dispersed, difficult access

Detecting Response Bias

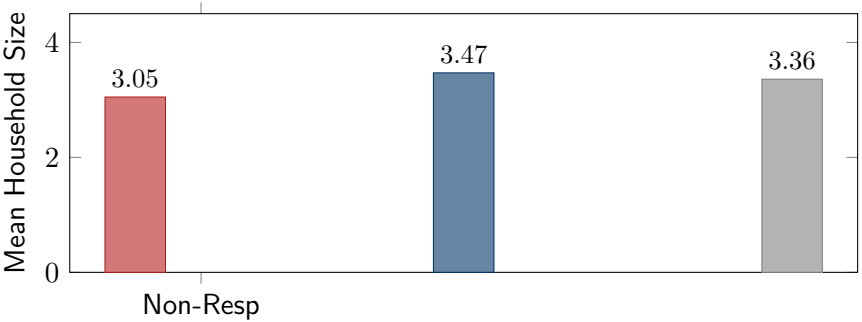
*** Key Insight**

Non-response creates bias **only if** respondents differ from non-respondents on the variables of interest.

Lindiwe's Test: Compare household size (available for all sampled units)

Group	Mean HH Size	SE
Respondents (n=15,839)	3.47	0.013
Non-Respondents (n=5,101)	3.05	0.024
Difference	**0.42**	—

Visualizing Response Bias



Bias Estimate: $(1 - 0.756) \times (3.47 - 3.05) = 0.244 \times 0.42 = \mathbf{0.10}$

If household size correlates with income, income estimates are also biased.

Measurement Error

Definition

Measurement error is the difference between the recorded value and the true value for a respondent, arising from the questionnaire, interviewer, respondent, or mode of data collection.

[Q]	[I]	[R]	[M]
Questionnaire	**Interviewer**	**Respondent**	**Mode**
Ambiguous wording, leading questions	Probing variation, recording errors	Recall failure, social desirability	CAPI vs PAPI differences

[Z] Zambara Case

Income questions trigger "social desirability bias" — respondents underreport high incomes and

Processing Error

Definition

Processing error arises during data capture, editing, coding, and transformation — including keying mistakes, coding inconsistencies, and software bugs.

Common Sources:

- Data entry errors (manual keying)
- OCR/scanning mistakes
- Coding inconsistencies
- Variable transformation errors
- Weighting calculation bugs
- Merging/matching errors
- Outlier handling decisions
- Software version differences

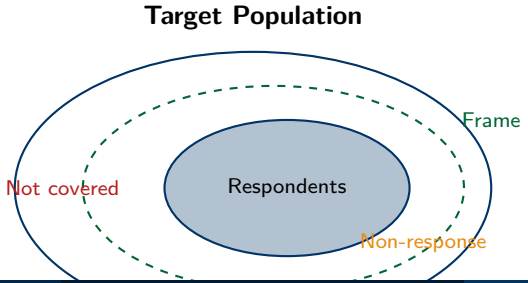
*** Key Insight**

Error Interactions: The Compound Problem

! Warning

Errors do not simply add up — they **interact and amplify** each other.

Example: Coverage × Non-Response Interaction



Diagnosing Zambara: TSE Mapping

[Z] Zambara Case

Lindiwe systematically maps her crises to the TSE framework to prioritize remediation efforts...

Error Type	Zambara Issue	Evidence
Coverage	12 PSUs inaccessible	Eastern Forests region
Sampling	Design effect = 1.34	Cluster homogeneity
Unit NR	RR = 75.6%	Below 85% target
Item NR	Income missing 24.4%	Sensitive question
Measurement	Income underreporting	Social desirability

Priority: Non-response (affects 24% of key variable) > Coverage (affects 0.6% of sample)

Knowledge Check

Time to Test Your Understanding!

?

Quiz Question 1

Question

A survey reports a mean income of R8,500 with a standard error of R150. Which error component does the standard error capture?

A) Total Survey Error
B) Sampling error only
C) Non-response error only
D) Coverage error only

Quiz Question 1

Question

A survey reports a mean income of R8,500 with a standard error of R150. Which error component does the standard error capture?

A) Total Survey Error
B) Sampling error only
C) Non-response error only
D) Coverage error only

*** Key Insight**

Quiz Question 2

Question

If respondents have larger households than non-respondents, and household size is positively correlated with income, what happens to income estimates?

A) Income is unbiased
B) Income is overestimated
C) Income is underestimated
D) Cannot determine from this information

Quiz Question 2

Question

If respondents have larger households than non-respondents, and household size is positively correlated with income, what happens to income estimates?

A) Income is unbiased
B) Income is overestimated
C) Income is underestimated
D) Cannot determine from this information

*** Key Insight**

Quiz Question 3

Question

Which of the following is a **measurement** error (not representation error)?

A) New informal settlements not in the sampling frame
B) Households refusing to participate
C) Respondent rounding their age to the nearest 5 years
D) Selected dwellings that no longer exist

Quiz Question 3

Question

Which of the following is a **measurement** error (not representation error)?

A) New informal settlements not in the sampling frame
B) Households refusing to participate
C) Respondent rounding their age to the nearest 5 years
D) Selected dwellings that no longer exist

* Key Insight

Answer: C) Respondent rounding their age

Quiz Question 4

Question

Doubling the sample size will approximately:

A) Halve all types of survey error
B) Reduce standard error by about 30%
C) Eliminate non-response bias
D) Reduce coverage error by half

Quiz Question 4

Question

Doubling the sample size will approximately:

A) Halve all types of survey error
B) Reduce standard error by about 30%
C) Eliminate non-response bias
D) Reduce coverage error by half

* Key Insight

Answer: B) Reduce standard error by about 30%

SE $\propto 1/\sqrt{n}$, so $\sqrt{2} \approx 1.41$ reduction $\approx 30\%$. Bias [illegible]

Quiz Question 5

Question

In Zambara, Northern Cape has the lowest response rate (68.4%). Using the non-response bias formula, which factor determines if this actually biases the national estimate?

A) The sample size of Northern Cape
B) Whether respondents differ from non-respondents
C) The number of interviewers deployed
D) The geographic area of the province

Quiz Question 5

Question

In Zambara, Northern Cape has the lowest response rate (68.4%). Using the non-response bias formula, which factor determines if this actually biases the national estimate?

A) The sample size of Northern Cape
B) Whether respondents differ from non-respondents
C) The number of interviewers deployed
D) The geographic area of the province

* Key Insight

Designing a Quality Monitoring Dashboard

[Z] Zambara Case

Lindiwe needs real-time monitoring to catch problems early. She designs a dashboard with key performance indicators (KPIs)...

Essential TSE Monitoring Indicators:

[RQ] Response Quality

- Response rate by domain
- Refusal vs non-contact ratio
- Item non-response rates
- Interview duration patterns

[DQ] Data Quality

- Range violations
- Skip logic failures
- Interviewer effects
- Digit preference (heaping)

Module 4.1 Summary

Key Takeaways:

1. TSE = Sampling + Non-Sampling errors
2. Non-sampling errors often exceed sampling error
3. Standard errors capture only the "tip of the iceberg"
4. Non-response bias depends on $(\bar{Y}_R - \bar{Y}_{NR})$
5. Systematic diagnosis enables targeted remediation

Zambara Status:

RR1	75.6%
Item NR (income)	24.4%
HH size bias	+0.42
Inaccessible PSUs	12

Next: Non-response adjustment strategies

References: Groves et al. (2009); Lohr (2022); AAPOR (2023)

Coming Up: Module 4.2

Non-Response Analysis & Diagnostics

Lab 4.2: Response Propensity Modeling

Build logistic models to predict non-response
Diagnose MAR vs MNAR mechanisms

[Z] Zambara Case

Lindiwe now understands the *what* and *where* of non-response. Next, she must understand the *why* — to design effective remediation.

Part II: Diagnosing Bias

The Field Dashboard Approach

Slides 31–60

Lindiwe Opens the Field Dashboard

[Z] Zambara Case

It's 8:15 AM. Lindiwe logs into the ZNSO Field Monitoring System. The dashboard loads with a cascade of status indicators. Her heart sinks as she sees more red than green...

ZAMBARA HLCS – FIELD MONITORING DASHBOARD

Response Rate	Item NR (Income)	PSUs Complete
75.6%	**24.4%**	**97.2%**
• Below Target	• Critical	• 12 Inaccessible

ALERTS: Northern Cape RR 68.4% | Eastern Forests 12 PSUs blocked | Income refusal spike

Designing Effective QA Dashboards

Dashboard Design Principles

A quality monitoring dashboard should provide **actionable intelligence** at a glance, not just data displays.

Essential Elements:

- Traffic light indicators
- Trend lines (not just snapshots)
- Drill-down capability
- Threshold alerts
- Comparative benchmarks

Key Questions Answered:

- Are we on target?
- Where are problems emerging?
- What's getting worse/better?
- Who needs intervention?
- What action is required?

Response Rate Key Performance Indicators

Response Rate Monitoring Panel

Indicator	Target	Current	Gap	Trend	Status
Overall RR1	$\geq$85%	75.6%	–9.4pp	$\downarrow$	●
Unit Refusal Rate	$\leq$10%	16.5%	+6.5pp	$\rightarrow$	●
Non-Contact Rate	$\leq$5%	7.9%	+2.9pp	$\downarrow$	●
Item NR (Income)	$\leq$5%	24.4%	+19.4pp	$\uparrow$	●
Callback Completion	$\geq$90%	82.3%	–7.7pp	$\uparrow$	●

* Key Insight

Traffic Light Thresholds:

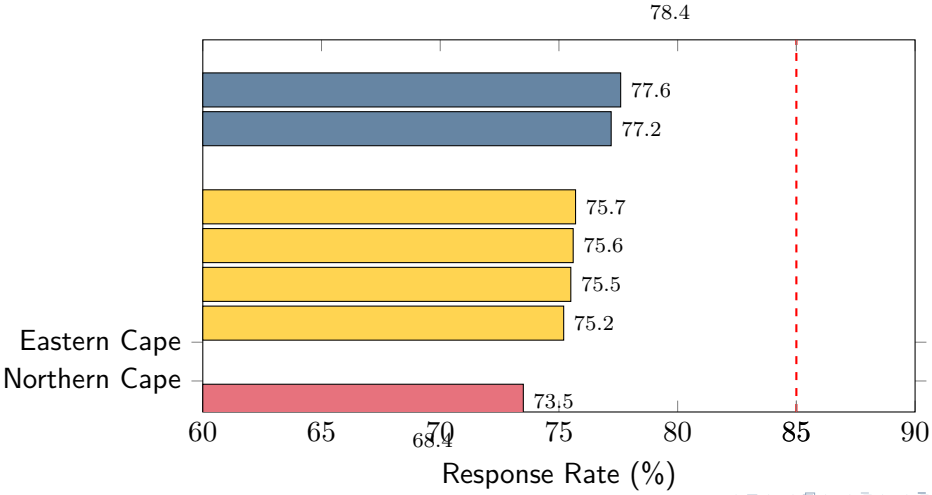
Provincial Response Rate Dashboard

Response Rates by Province – Lab 4.1 Results

Province	Sample	Complete	RR1%	Refusal%	NC%	Status
Northern Cape	855	585	**68.4**	21.5	10.1	● Critical
Eastern Cape	2,706	1,989	**73.5**	18.4	8.1	● Critical
Gauteng	5,526	4,156	75.2	16.7	8.1	● Watch
Mpumalanga	1,671	1,262	75.5	16.6	7.8	● Watch
Western Cape	1,931	1,460	75.6	15.7	8.7	● Watch
Free State	1,276	966	75.7	16.7	7.6	● Watch
KwaZulu-Natal	3,257	2,516	77.2	15.3	7.5	● Watch
North West	1,333	1,035	77.6	15.0	7.4	● Watch
Limpopo	2,385	1,870	**78.4**	14.8	6.8	● Best
NATIONAL	**20,940**	**15,839**	**75.6**	**16.5**	**7.9**	●

Source: Lab 4.1 Simulation Output

Provincial Response Rates: Visual Comparison



Understanding Geographic Variation

Why does Northern Cape perform worst?

- Vast geographic area (largest province)
- Sparse population distribution
- Long travel distances between PSUs
- Limited interviewer pool
- Mining area privacy concerns

Why does Limpopo perform best?

- Strong community cooperation
- Traditional leadership support
- Experienced field teams

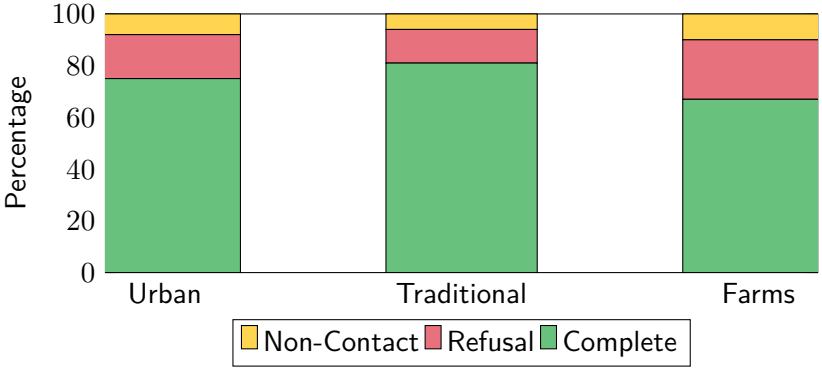
Gap Analysis

Spread: 10.0 percentage points (68.4% to 78.4%)

Below 75%: 2 provinces
Below 80%: 9 provinces
At target: 0 provinces

- **No province meets 85% target**

Response Patterns by Settlement Type



Urban: High refusals
Privacy, gatekeeping

Traditional: Best response
Community trust

Farms: High non-contact
Access challenges

Drilling Down: Province × Geography Type

Response Rate Heat Map (%)

Province	Urban	Traditional	Farms	Overall
Northern Cape	65.2	74.8	61.3	68.4
Eastern Cape	70.1	76.2	68.5	73.5
Gauteng	74.8	82.1	69.2	75.2
KwaZulu-Natal	75.5	80.4	71.8	77.2
Limpopo	76.2	81.5	73.4	78.4
Overall	**75.0**	**81.0**	**67.0**	**75.6**

! Warning

The Critical Question: Does Non-Response Create Bias?

The Bias Condition

Non-response creates bias **if and only if** respondents differ systematically from non-respondents on variables of interest.

Non-Response Bias Formula:

$$\boxed{Bias_{NR} = (1 - RR) \times (\bar{Y}_R - \bar{Y}_{NR})}$$

Two Components:

1. $(1 - RR)$: Non-response rate
2. $(\bar{Y}_R - \bar{Y}_{NR})$: Difference in means

Key Insight:

- High NR + no difference = No bias
- Low NR + large difference = Some bias
- Both high = Severe bias

Testing for Response Bias

[Z] Zambara Case

Lindiwe cannot directly observe values for non-respondents. But she *can* compare respondents and non-respondents on **auxiliary variables** available for all sampled units.

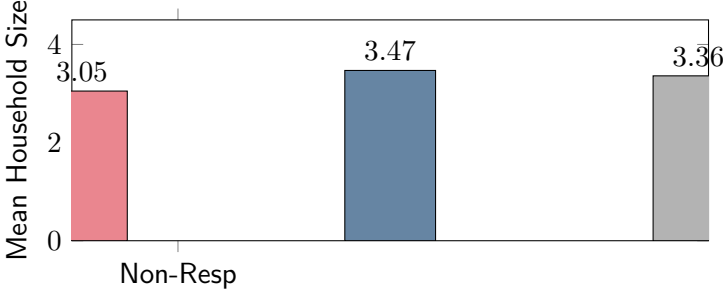
Available Auxiliary Variables:

- **Household size** — observed for all sampled dwellings
- **Dwelling type** — from frame/observation
- **Geographic location** — known by design
- **Previous survey response** — from panel if applicable

* Key Insight

Lab 4.1 Results: Household Size Comparison

Response Bias Diagnostic – Household Size



Difference: $3.47 - 3.05 = \mathbf{0.42}$ **persons** **t-statistic:** 15.8 **p-value:** < 0.001

Interpreting the Bias Evidence

! Warning

Statistically Significant Difference Detected!
Non-respondents have systematically **smaller households** than respondents.

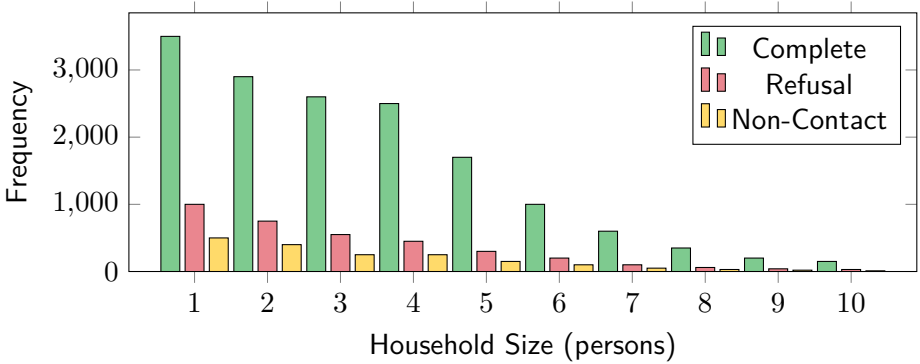
Calculating Non-Response Bias:

$$
\begin{aligned}
Bias_{NR} &= (1 - RR) \times (\bar{Y}_R - \bar{Y}_{NR}) \\
&= (1 - 0.756) \times (3.47 - 3.05) \\
&= 0.244 \times 0.42 = \mathbf{0.10}
\end{aligned}
$$

Interpretation:

- Mean household size is **overestimated** by 0.10 persons

Visualizing the Distribution Shift



*** Key Insight**

Higher proportion of size-1 households among non-respondents — harder to contact and

Implications for Key Survey Indicators

[Z] Zambara Case

If small households are underrepresented, what does this mean for Lindiwe's poverty estimates and other key indicators?

Likely Overestimated:

- Mean household income
- Per capita consumption
- Asset ownership rates
- Housing quality indices

(Larger HH tend to have more resources)

Likely Underestimated:

- Poverty headcount rate
- Elderly living alone
- Single-parent households
- Youth unemployment

(Vulnerable groups often in smaller HH)

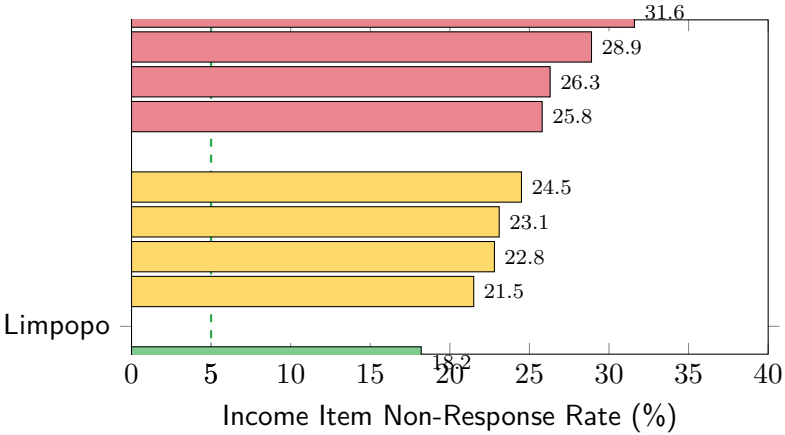
Item Non-Response: The Income Problem

Item Non-Response Rates by Variable

Variable	Asked	Missing	NR Rate	Status
Household size	15,839	12	0.1%	●
Dwelling type	15,839	45	0.3%	●
Water source	15,839	89	0.6%	●
Electricity	15,839	67	0.4%	●
Education (head)	15,839	234	1.5%	●
Employment status	15,839	512	3.2%	●
Total HH income	**15,839**	**3,865**	**24.4%**	● **Critical**

*** Key Insight**

Income Non-Response: Geographic Patterns



Dashed green line = 5% acceptable threshold

Why Is Income Data Missing?

Reason Codes – Income NR

Reason	%
"Refused to answer"	42.3
"Don't know"	28.7
"Irregular/variable"	15.2
"Confidential"	8.9
Other/blank	4.9

Interpretation:

- **42% refusals** — sensitivity
- **29% don't know** — complex finances
- **15% irregular** — informal economy
- Higher in urban, mining areas

! Warning

Income non-response is **NOT random** — concentrated among both the very wealthy

Missing Data Mechanisms

Little & Rubin Classification

The mechanism generating missing data determines which remediation strategies are valid.

MCAR
Missing Completely At Random

Missingness unrelated to any data

✓ Complete case OK

MAR
Missing At Random

Missingness depends on observed data

∼ Imputation OK

MNAR
Missing Not At Random

Missingness depends on missing values

× Problematic

[Z] Zambara Case

Testing the MAR Assumption

Diagnostic: Compare characteristics of income respondents vs. non-respondents.

Income Response vs. Auxiliary Variables

Characteristic	Responded	Refused	p-value
Mean household size	3.52	3.18	<0.001
% Urban	58.2	71.4	<0.001
% with electricity	89.1	94.6	<0.001
% owns vehicle	23.4	31.8	<0.001
Mean rooms in dwelling	4.2	5.1	<0.001

*** Key Insight**

Crisis 2: The Inaccessible PSUs

[Z] Zambara Case

Lindiwe receives an urgent message: "Security situation deteriorating. 12 PSUs now completely inaccessible. Advise on protocol."

Inaccessible PSU Summary

District	PSUs	Planned HH	Status
Eastern Forests North	5	580	No access – civil unrest
Eastern Forests South	4	465	No access – road blocked
Border Zone	3	355	Security advisory
Total	**12**	**1,400**	—

Coverage Bias from Inaccessible Areas

The Affected Areas Are Not Random:

Eastern Forests Profile:

- 78% traditional settlements
- Agriculture-dominant economy
- Below-average income levels
- Higher poverty rates
- Limited infrastructure

National Average:

- 28% traditional settlements
- Mixed economy
- Average income levels
- Average poverty rates
- Better infrastructure

! Warning

Dropping these PSUs will **underrepresent poor, rural households** and likely **underestimate national poverty rates**.

Options for Handling Inaccessible PSUs

Option A: Drop PSUs

Pros: Simple, no assumptions
Cons: Coverage bias, reduced sample, violates design

Option B: Substitute PSUs

Pros: Maintains sample size, can match characteristics
Cons: Substitutes may differ, requires reserve list

Option C: Reweight

Pros: Uses existing data
Cons: Assumes accessible PSUs can represent inaccessible ones

* Key Insight

Best practice: Document decision and conduct **sensitivity analysis** comparing results under different approaches.

Activity: Think-Pair-Share

Discussion Question

You are advising Lindiwe on the Eastern Forests situation:

Which option would you recommend and why?

Consider: bias implications, feasibility, documentation

Instructions:

1. **Think** (2 min): Consider your recommendation individually
2. **Pair** (3 min): Discuss with your neighbor
3. **Share** (5 min): Report key points to the group

Zambara Quality Summary Dashboard

ZAMBARA HLCS – QUALITY SUMMARY

Unit Non-Response

RR1: 75.6%
Target: 85%
Gap: −9.4pp

Item NR (Income)

NR Rate: 24.4%
Target: 5%
Gap: +19.4pp

Coverage

PSUs: 97.2%
12 Inaccessible
Bias Risk: Mod

Response Bias Evidence

HH Size Diff:	0.42 ($p<0.001$)
Direction:	Small HH under-rep
Rel. Bias:	3.0%

ACTIONS REQUIRED

1. Non-response weight adjustment
2. Income imputation strategy
3. PSU substitution protocol
4. Bias documentation

Lab 4.1 Exercise: Key Findings Review

Lab 4.1 Output Summary

Task: Analyze response rates and diagnose bias using GHS 2024 data

Key Results:

- Simulated AAPOR RR1: 75.6% (target: 85%)
- Provincial range: 68.4% (Northern Cape) to 78.4% (Limpopo)
- Geographic pattern: Farms worst (67%), Traditional best (81%)
- Response bias: HH size difference of 0.42 ($t=15.8$, $p<0.001$)
- Direction: Small households systematically underrepresented

Skills Practiced: Response status creation, AAPOR RR calculation, bias testing, visualization

Lab 4.1: R Code Highlights

Key R Functions Used

```
# Response rates by province
rr_by_province <- ghs_hh %>%
  group_by(prov_name) %>%
  summarise(
    n_total = n(),
    n_complete = sum(response_status == 1),
    rr1 = n_complete / n_total * 100
  )

# T-test for bias detection
t.test(respondents$hholdsz, nonrespondents$hholdsz)
```

Lab 4.1: STATA Code Highlights

Key STATA Commands Used

```
* Response rates by province
collapse (count) n_total=hhid ///
  (sum) n_complete = (response_status == 1), by(prov)
gen rr1 = n_complete / n_total * 100

* T-test for bias detection
ttest hholdsz, by(respondent)
```

Module 4.1: Key Takeaways

1. **TSE Framework** provides systematic diagnosis of all error sources
2. **Non-sampling errors** often exceed sampling error but are hidden
3. **Response bias** $= (1 - RR) \times (\bar{Y}_R - \bar{Y}_{NR})$
4. **Dashboard monitoring** enables early detection and intervention
5. **Auxiliary variables** reveal bias even when Y is unobserved
6. **Geographic patterns** help target remediation efforts
7. **Documentation** of bias direction is essential for users

*** Key Insight**

The goal is not to eliminate all error, but to **understand, quantify, and communicate** the error to data users.

Item Non-Response Strategies

The Item Non-Response Problem

Lindiwe's income variable has 26.4% missing values.

Status	**Count**
Valid income	15,411
Missing income	5,529
Total	**20,940**

What should she do with the missing 26.4%?

Three Common Approaches

Option 1: Listwise Deletion — Drop cases with missing values

Option 2: Mean Imputation — Fill with overall mean

Option 3: Hot-Deck Imputation — Fill with donor values

Which is best? Let's compare using Lab 4.2 results.

Option 1: Listwise Deletion (Complete Case Analysis)

Method: Analyze only cases with complete data.

	Full Data	**Complete Cases**
N	20,940	15,411
Lost	—	5,529 (26.4%)

Problem: Over a quarter of data discarded!

The Cost of Listwise Deletion

What you lose:

- 5,529 households of information
- Statistical power
- Precision of estimates
- Money spent collecting that data

Hidden danger: If missingness is not MCAR, estimates are biased.

Complete Case Analysis: Bias Check

Statistic	True	Complete Cases
Mean (R)	218,305	214,421
Median (R)	5,257	5,180
Std Dev (R)	1,421,646	1,409,264
Mean Bias	—	–1.78%

Even "just dropping" missing cases introduces bias.

When Is Listwise Deletion Acceptable?

Only when:

- Missing is truly MCAR (rare)
- Missing rate is very low ($<5\%$)
- You can afford to lose sample size

At 26.4% missing, deletion is **not acceptable**.

Option 2: Mean Imputation

Method: Replace all missing values with the mean.

Item	**Value**
Observed Mean	R 214,421
Missing Values	5,529
All imputed as	**R 214,421**

Every missing case gets the *same* value.

The Tragedy of Mean Imputation

5,529 households

all assigned income of

R 214,421

All variability in those 5,529 cases is **destroyed**.

Mean Imputation: Variance Shrinkage

Statistic	True	Mean Imputed
Mean (R)	218,305	214,421
Std Dev (R)	1,421,646	1,208,970
SD Change	—	**–15%**

Variance underestimated by 15%!

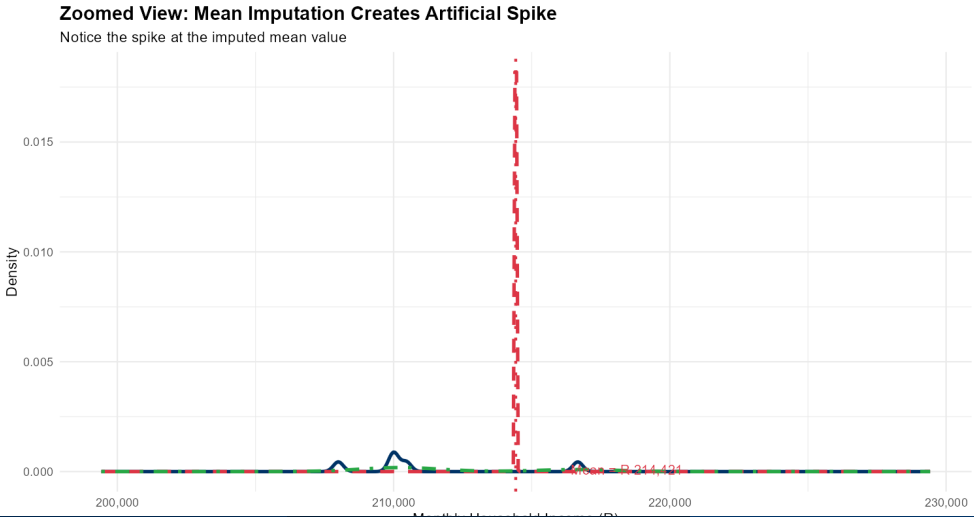
Why Does Variance Shrink?

Before imputation:
Income values: 1,000 . . . 50,000 . . . 200,000 . . . 1,000,000

After mean imputation:
Income values: 1,000 . . . 214,421 . . . 214,421 . . . 1,000,000

5,529 values clustered at single point → variance shrinks.

Visual: The Artificial Spike



Mean Imputation: Median Distortion

Statistic	True	Mean Imputed
Median (R)	5,257	8,960
Change	—	**+70%**

The median is **dragged upward** by 5,529 identical high values.

Mean Imputation: Summary of Problems

1. Underestimates variance (-15%)
2. Creates artificial spike in distribution
3. Distorts median ($+70\%$)
4. Attenuates correlations with other variables
5. Produces artificially narrow confidence intervals

Never use mean imputation for continuous variables.

Option 3: Hot-Deck Imputation

Method: Replace missing with values from similar "donor" cases.

Key features:

- Imputed values are *real* observed values
- Donors selected from same "class"
- Preserves distribution shape

Defining Donor Classes

Lab 4.2 Classes: Province $\times$ Household Size Group

Dimension	**Categories**
Province	9 provinces
HH Size	1, 2–3, 4–5, 6+
Total classes	**36**

Donors matched on geography and household composition.

Hot-Deck: The Process

For each missing case:

1. Identify its imputation class
2. Find all donors (observed) in that class
3. Randomly select one donor
4. Copy donor's income value

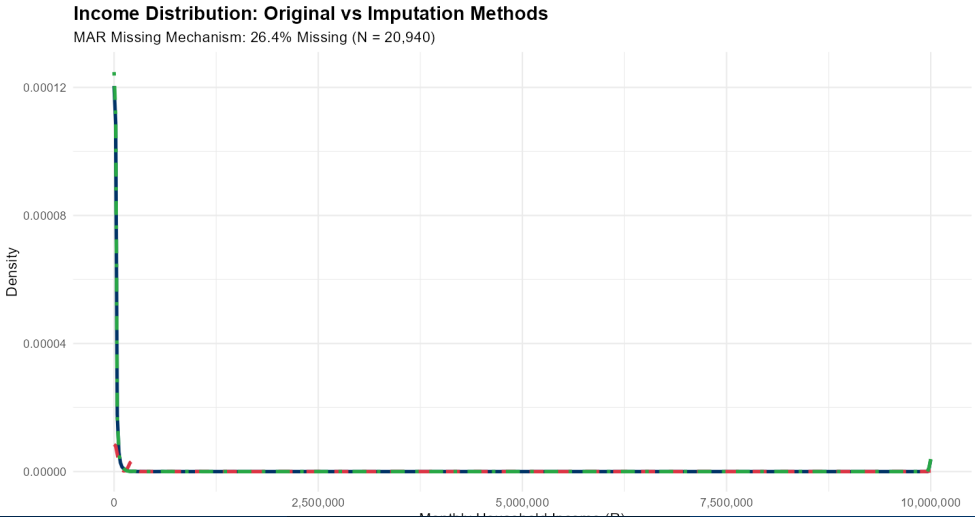
Result: Imputed values have **natural variability**.

Hot-Deck: Lab 4.2 Results

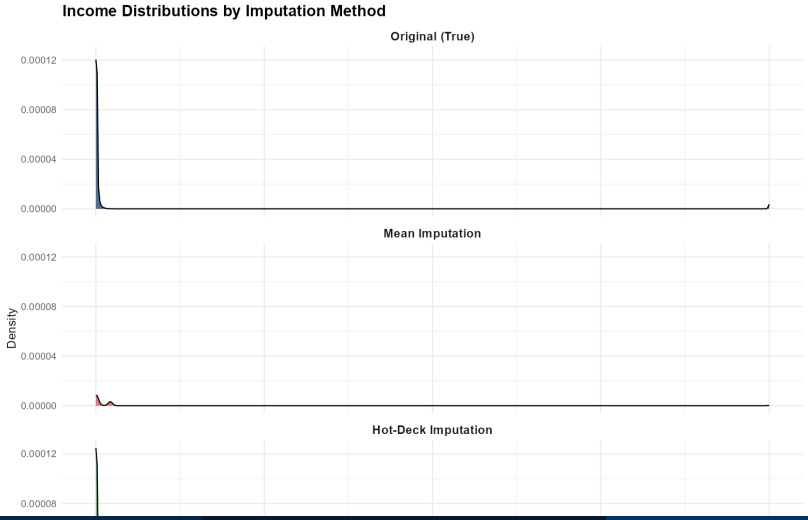
Statistic	True	Hot-Deck
Mean (R)	218,305	223,244
Median (R)	5,257	5,180
Std Dev (R)	1,421,646	1,439,248
Mean Bias	—	+2.26%
SD Change	—	**+1.2%**

Variance preserved! (vs –15% for mean imputation)

Distribution Comparison



Faceted View: Three Distributions



Head-to-Head: Mean vs Hot-Deck

Metric	Mean Imp.	Hot-Deck
Mean Bias	−1.78%	+2.26%
Median Bias	+70%	−1.5%
SD Change	**−15%**	**+1.2%**
Distribution	Distorted	Preserved

Winner: Hot-Deck on all distribution metrics.

Why Hot-Deck Preserves Variance

Mean Imputation:
5,529 missing $\rightarrow$ all get value 214,421

Hot-Deck:
5,529 missing $\rightarrow$ get 5,529 *different* real values

Hot-deck maintains the **spread** of the distribution.

Comparing Imputed Values Only

	Mean Imp.	**Hot-Deck**
N imputed	5,529	5,529
Mean of imputed	R 214,421	R 218,XXX
SD of imputed	**0**	R 1.4M

Mean imputation: SD = 0 (all identical!)
Hot-deck: SD matches original distribution.

Impact on Correlations

True correlation (Income $\times$ HH Size): $\rho = 0.XX$

After Mean Imputation: Correlation attenuated

After Hot-Deck: Correlation preserved

Mean imputation weakens relationships between variables.

Impact on Confidence Intervals

Mean Imputation:

- Underestimates variance
- SE too small
- CI too narrow
- False precision!

Hot-Deck:

- Preserves variance
- Appropriate SE
- Honest uncertainty

Full Method Comparison

Method	N	Mean Bias	SD Change	Verdict
Listwise Deletion	15,411	–1.78%	–0.9%	Data loss
Mean Imputation	20,940	–1.78%	–15%	Distorted
Hot-Deck	20,940	+2.26%	+1.2%	Best

Choosing an Imputation Method

Continuous variables (income, expenditure):
→ Hot-Deck or Regression Imputation

Categorical variables (education, employment):
→ Hot-Deck or Logistic Imputation

Never acceptable:
→ Mean/Mode imputation for substantive analysis

Beyond Single Imputation

Limitation of Hot-Deck: Treats imputed values as "real."

Solution: Multiple Imputation

- Create M imputed datasets (typically 5–20)
- Analyze each separately
- Combine results using Rubin's rules
- Properly accounts for imputation uncertainty

Advanced topic for Day 5.

Module 4.2: Key Takeaways

1. Listwise deletion loses data and may bias estimates
2. Mean imputation **destroys variance** (–15% in Lab 4.2)
3. Hot-deck **preserves distribution** (+1.2% SD change)
4. Choose imputation classes based on predictive variables
5. Document your imputation method in metadata

Next: Module 4.3 — Adaptive Survey Design

From Item Non-Response to Unit Non-Response

4.2: Missing values within cases

$\downarrow$

4.3: Missing entire cases (PSUs)

Short Break: 10 minutes

Mastering Hot-Deck Imputation

The Goal: Fill 5,529 Missing Income Values

Lindiwe needs to impute 26.4% missing income data.

Requirements:

- Preserve distribution shape
- Maintain variance
- Use realistic values

Solution: Hot-Deck Imputation

Hot-Deck: The Core Concept

Idea: Borrow values from similar *observed* cases.

Recipient:	Case with missing value
Donor:	Similar case with observed value
Donation:	Copy donor's value to recipient

Key: Define "similar" using **imputation classes**.

Step 1: Define Imputation Classes

Group records into homogeneous cells

based on variables that **predict** income.

Step 1: Lab 4.2 Imputation Classes

Variables used:

- Province (9 categories)
- Household size group (4 categories)

HH Size Group	Definition
1	Single person
2–3	Small household
4–5	Medium household
6+	Large household

Total classes: $9 \times 4 = 36$

Why Province and HH Size?

Province: Income varies geographically

HH Size: Larger HH = more earners = higher income

Principle: Classes should be:

- Predictive of the missing variable
- Large enough to have donors
- Homogeneous within cells

Step 2: Identify Donors and Recipients

Within each class, separate:

Donors = Cases with observed income
Recipients = Cases with missing income

Step 2: Example — Gauteng, HH Size 2–3

ID	Class	Income	Role
1001	GP_2-3	R 8,500	Donor
1002	GP_2-3	R 12,200	Donor
1003	GP_2-3	?	Recipient
1004	GP_2-3	R 9,800	Donor
1005	GP_2-3	?	Recipient
1006	GP_2-3	R 15,400	Donor

4 donors available for 2 recipients.

Step 3: Random Donor Selection

For each recipient:

Randomly select ONE donor from the same class.

Copy the donor's income value.

Step 3: Selecting a Donor (1/3)

Recipient: ID 1003 (Gauteng, HH Size 2–3)

Available Donors:

- ID 1001: R 8,500
- ID 1002: R 12,200
- ID 1004: R 9,800
- ID 1006: R 15,400

Action: Random selection...

Step 3: Selecting a Donor (2/3)

Recipient: ID 1003 (Gauteng, HH Size 2–3)

Random number generated: 0.67

Selected donor: ID 1004

Donated value: **R 9,800**

Step 3: Selecting a Donor (3/3)

Result:

ID	Class	Before	After
1003	GP_2-3	?	**R 9,800**

ID 1003 now has a *real* income value from a similar household.

Step 4: Repeat for All Recipients

Process all 5,529 missing cases

Each gets a value from a donor in same class.

Different recipients may get different donors.

Complete Example: Before and After

ID	Province	HH Size	Before	After
1001	Gauteng	3	R 8,500	R 8,500
1002	Gauteng	2	R 12,200	R 12,200
1003	Gauteng	3	?	R 9,800
1004	Gauteng	2	R 9,800	R 9,800
1005	KZN	5	?	R 4,200
1006	KZN	4	R 4,200	R 4,200

1003 $\leftarrow$ Donor 1004 (same class)
1005 $\leftarrow$ Donor 1006 (same class)

Why Random Selection?

Q: Why not always use the "closest" donor?

A: Random selection:

- Preserves within-class variance
- Avoids artificial clustering
- Enables variance estimation via replication

R Code: Creating Imputation Classes

```
# Create household size groups
ghs$hhsize_grp <- cut(ghs$hholdsz,
                      breaks = c(0, 1, 3, 5, Inf),
                      labels = c("1", "2-3", "4-5", "6+"))

# Create imputation class
ghs$imp_class <- paste(ghs$prov,
                       ghs$hhsize_grp,
                       sep = "_")
```

R Code: Performing Hot-Deck

```
# For each missing case
for (i in missing_idx) {
  this_class <- ghs$imp_class[i]

  # Find donors in same class
  donors <- ghs$income[ghs$imp_class == this_class
                       & !is.na(ghs$income)]

  # Random selection
  ghs$income_imputed[i] <- sample(donors, 1)
}
```

STATA Code: Hot-Deck

```
* Create imputation class
egen imp_class = group(prov hhsize_grp)

* Hot-deck imputation
* (using hotdeck package)
hotdeck totmhinc, by(imp_class) ///
    store(income_imputed)
```

Lab 4.2: Imputation Class Summary

Item	Value
Total classes	36
Total recipients	5,529
Total donors	15,411
Mean donors per class	428
Min donors in a class	12

Adequate donor pool in all classes.

The Key Question

Did Hot-Deck preserve the income distribution?

Compare: Original vs Hot-Deck Imputed

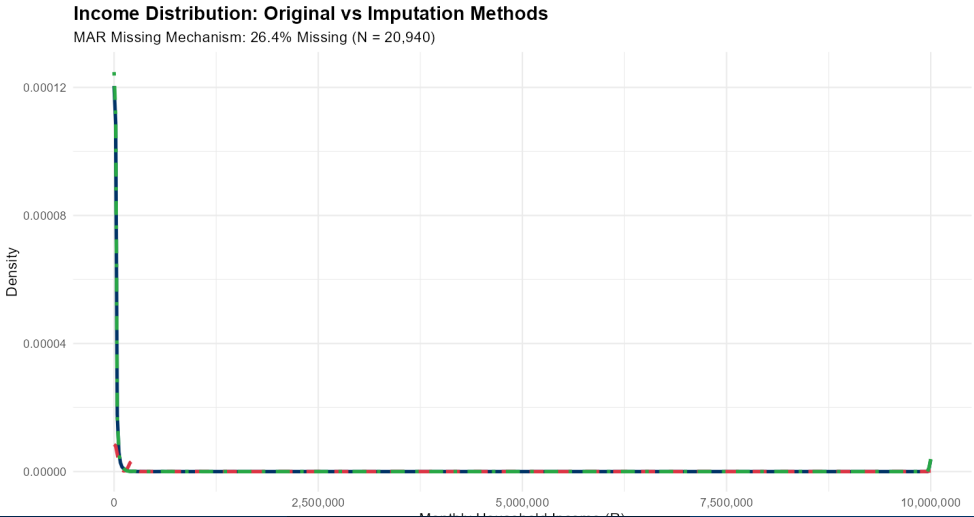
Test: Density plots and summary statistics.

Lab 4.2: Distribution Comparison

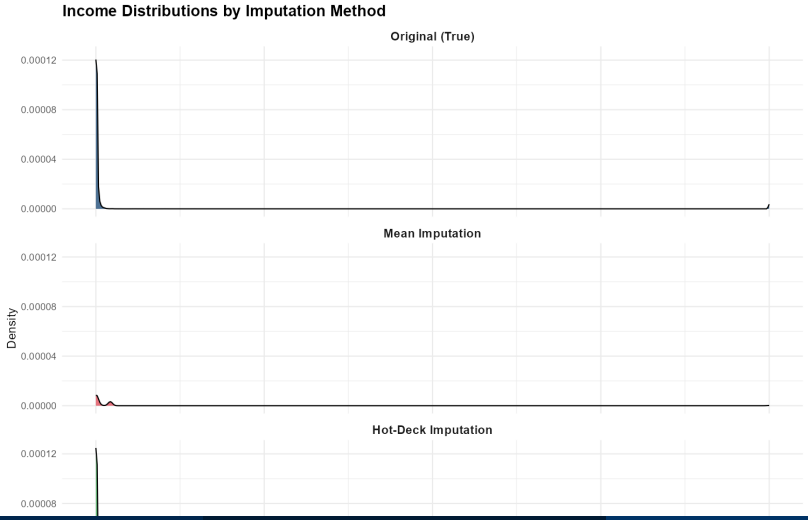
Statistic	Original	Hot-Deck
N	20,940	20,940
Mean (R)	218,305	223,244
Median (R)	5,257	5,180
Std Dev (R)	1,421,646	1,439,248
Mean change	—	+2.3%
SD change	—	**+1.2%**

Distribution preserved!

Before vs After: Density Overlay



Distribution Shape Preserved

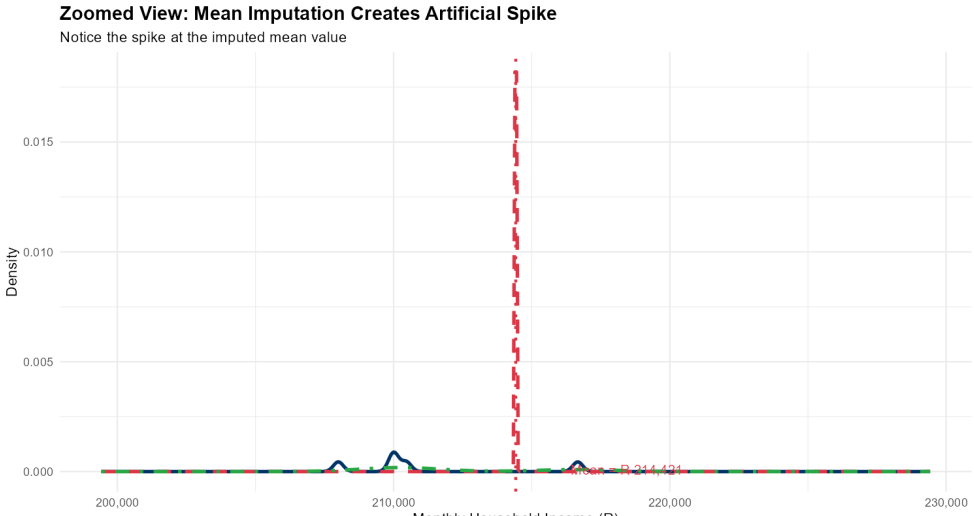


Contrast: Hot-Deck vs Mean Imputation

Metric	Mean Imp	Hot-Deck
SD Change	−15%	+1.2%
Median Change	+70%	−1.5%
Distribution	Distorted	Preserved

Hot-Deck wins on every distribution metric.

Zoomed: The Mean Imputation Spike



Practical Issue: Small Cells

Problem: Some classes may have few donors.

Class	Recipients	Donors
NC_1	8	3
NC_6+	5	2

Risk: Same donor used multiple times.

Solution: Cell Collapsing

When a class is too small:

1. Merge with similar class
2. Remove least predictive variable
3. Use hierarchical collapsing

Example:
NC_1 + NC_2-3 → NC_Small

Hot-Deck Variants

Random within class: Our method (simple, fast)

Sequential: Use nearest previous donor in sorted file

Nearest neighbor: Minimize distance metric

Weighted: Weight donors by similarity score

All preserve distribution better than mean imputation.

Hot-Deck: Key Takeaways

1. Define classes using **predictive variables**
2. Impute from **real observed values**
3. **Random selection** preserves variance
4. **Validate** with density comparison
5. Handle small cells via **collapsing**

Lab 4.2 Result: SD change only **+1.2%**

Next: Adaptive Design in Conflict Zones

From Item Non-Response to Unit Non-Response

Hot-Deck handles missing *values*.

What about missing *entire PSUs*?

→ Module 4.3: Adaptive Survey Design

Adaptive Survey Design

Sampling in Conflict Zones

Monday Morning: The Security Alert

Lindiwe's phone buzzes at 7:15 AM...

ZNSO SECURITY ALERT

Field operations suspended in Eastern Highlands.
50% of PSUs now inaccessible.
Await further instructions.

The Situation: Eastern Highlands Crisis

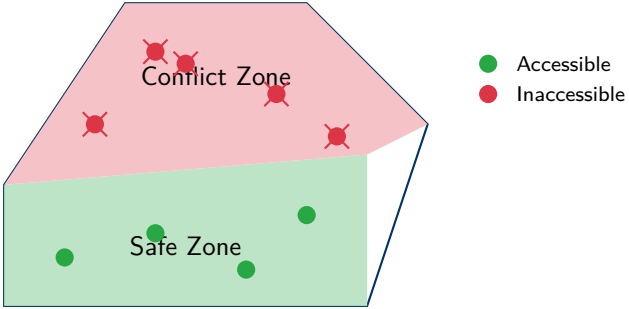
Affected Province: Eastern Highlands (KwaZulu-Natal)

Impact:

- 50% of provincial PSUs inaccessible
- 1,539 planned households affected
- Field teams cannot enter safely

Question: What should Lindiwe do?

Eastern Highlands: Affected Areas



Scale of the Problem

Item	Value
Total sample (national)	20,940
Eastern Highlands sample	3,078
PSUs inaccessible	50%
Households lost	1,539
National impact	**7.3%**

Not catastrophic nationally, but **devastating** for provincial estimates.

Lindiwe's Options

Strategy A: Drop inaccessible PSUs

Strategy B: Re-weight remaining PSUs

Strategy C: Substitute with "safe" neighbor PSUs

Which strategy minimizes bias?

Strategy A: Drop Inaccessible PSUs

Method: Simply exclude inaccessible areas from analysis.

Advantages:

- Simplest approach
- No assumptions required
- Quick to implement

Disadvantages:

- Loses sample size
- Coverage bias if areas differ

Strategy A: What Happens?

	Full Sample	**After Drop**
Households	20,940	19,401
Lost	—	1,539 (7.3%)

The dropped PSUs are **not random** — they're in conflict areas.

These areas may have different characteristics.

Strategy A: The Bias

Estimate	True	Strategy A
Mean Income (R)	260,166	244,458
Bias (R)	—	–15,708
Relative Bias	—	**–6.04%**

Dropping PSUs **underestimates** income by 6%.

Conflict areas had *higher* average income.

Strategy B: Re-weight Remaining PSUs

Method: Inflate weights of accessible PSUs to compensate.

Logic: Accessible PSUs "represent" inaccessible ones.

Weight adjustment:

$$w_{adj} = w_{orig} \times \frac{\sum w_{stratum}}{\sum w_{accessible}}$$

Strategy B: Weight Adjustment

For each affected stratum:

Item	**Value**
Original weight sum	1,000,000
Accessible weight sum	500,000
Adjustment factor	**2.0**

Remaining PSUs get **double weight**.

Strategy B: The Critical Assumption

Accessible PSUs are **representative** of inaccessible ones.

If true: Re-weighting eliminates bias.

If false: Re-weighting may amplify bias.

Is this assumption reasonable in a conflict zone?

Strategy B: The Bias

Estimate	True	Strategy B
Mean Income (R)	260,166	268,100
Bias (R)	—	+7,934
Relative Bias	—	**+3.05%**

Re-weighting **overestimates** income by 3%.

Better than dropping (3% vs 6%), but still biased.

Strategy C: Substitute with Neighbor PSUs

Method: Replace inaccessible PSUs with "similar" accessible ones.

Process:

1. For each inaccessible PSU
2. Find accessible PSU in same stratum
3. Duplicate its data as substitute

Strategy C: Finding Substitutes

Matching criteria:

- Same stratum (province $\times$ geotype)
- Similar PSU size
- Geographic proximity (when possible)

Lab 4.3: Random selection within stratum.

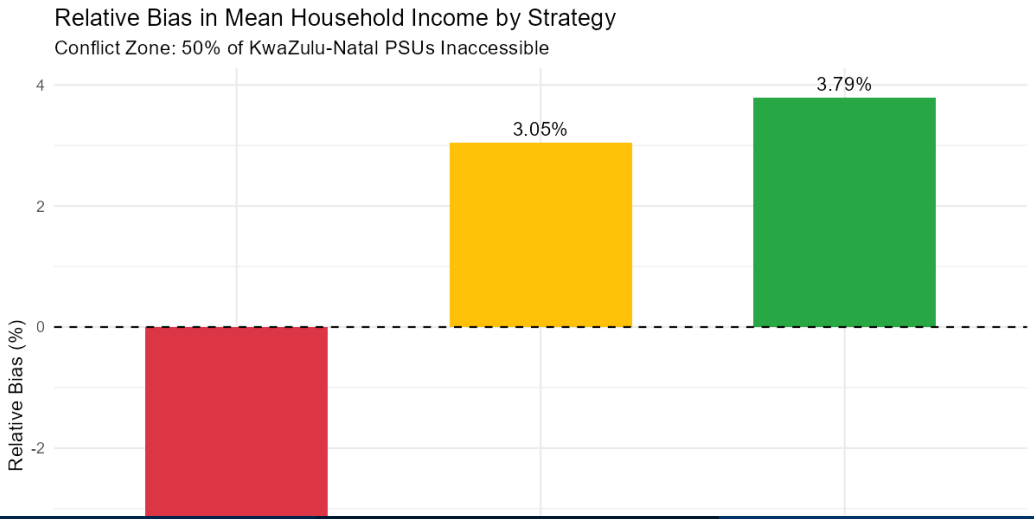
Strategy C: The Bias

Estimate	True	Strategy C
Mean Income (R)	260,166	270,020
Bias (R)	—	+9,854
Relative Bias	—	**+3.79%**

Substitution **overestimates** by 3.8%.

Similar to re-weighting, but maintains sample size.

Strategy Comparison: Visual



Full Strategy Comparison

Strategy	N	Income (R)	Bias	RelBias
True (Full)	20,940	260,166	0	0%
A: Drop	19,401	244,458	−15,708	**−6.04%**
B: Re-weight	19,401	268,100	+7,934	+3.05%
C: Substitute	20,955	270,020	+9,854	+3.79%

Winner by absolute bias: Strategy B (Re-weight) at 3.05%

Why Is Dropping the Worst?

The conflict zone is not random.

Eastern Highlands conflict areas have:

- Higher average income
- More urban settlements
- Different economic profile

Dropping them **systematically excludes** higher-income areas.

Understanding Bias Direction

Strategy	Direction	Why
A: Drop	Underestimate	Excluded high-income areas
B: Re-weight	Overestimate	Safe areas have higher income
C: Substitute	Overestimate	Substitutes from safe areas

All strategies have bias — the question is **how much**.

Real-World Considerations

Beyond bias:

- **Variance:** Re-weighting inflates standard errors
- **Feasibility:** Substitution requires field logistics
- **Documentation:** All approaches need transparency
- **Ethics:** Safety of field staff is paramount

When to Use Each Strategy

Drop (A): When inaccessible areas are similar to accessible.

Re-weight (B): When PSUs within strata are exchangeable.

Substitute (C): When good matching is possible.

Best practice: Combine approaches + sensitivity analysis.

Always: Sensitivity Analysis

Report multiple scenarios:

Scenario	Estimate
Base case (re-weight)	R 268,100
Lower bound (drop)	R 244,458
Upper bound (substitute)	R 270,020

Users understand the **range of uncertainty**.

Documentation Requirements

Always document:

1. Number and location of inaccessible PSUs
2. Reason for inaccessibility
3. Strategy chosen and rationale
4. Impact on estimates (bias analysis)
5. Limitations and caveats

Lab 4.3: R Code Overview

```
# Strategy A: Drop
ghs_A <- ghs %>% filter(accessible == TRUE)

# Strategy B: Re-weight
ghs_B <- ghs_A %>%
  mutate(wgt_adj = house_wgt * wgt_factor)

# Strategy C: Substitute
# Duplicate donor PSU data for recipients
```

Lab 4.3: STATA Code Overview

```
* Strategy A: Drop
keep if accessible == 1

* Strategy B: Re-weight
gen house_wgt_adj = house_wgt * wgt_factor

* Strategy C: Substitute
* Append donor PSU records
```

Discussion: Your Experience

Have you faced inaccessible areas in your surveys?

Consider:

- What caused the inaccessibility?
- What strategy did you use?
- How did you document it?

Small group discussion: 5 minutes

Module 4.3: Key Takeaways

1. Inaccessible PSUs create **coverage bias**
2. Dropping is simple but often **worst** for bias
3. Re-weighting assumes **exchangeability**
4. Substitution maintains n but needs **good matching**
5. Always do **sensitivity analysis**
6. **Document** everything

Lab 4.3: What We Learned

Scenario: 50% of Eastern Highlands PSUs inaccessible.

Strategy	Absolute Bias
A: Drop	6.04%
B: Re-weight	**3.05%**
C: Substitute	3.79%

Best performer: Re-weighting (lowest absolute bias).

Next: Module 4.4 — Quality Assurance Integration

From Crisis Response to Prevention

4.3: Handling problems when they occur

↓

4.4: Building quality into the process

Lunch Break: 60 minutes

Adaptive Protocols

Decision Frameworks for Field Challenges

The Decision Problem

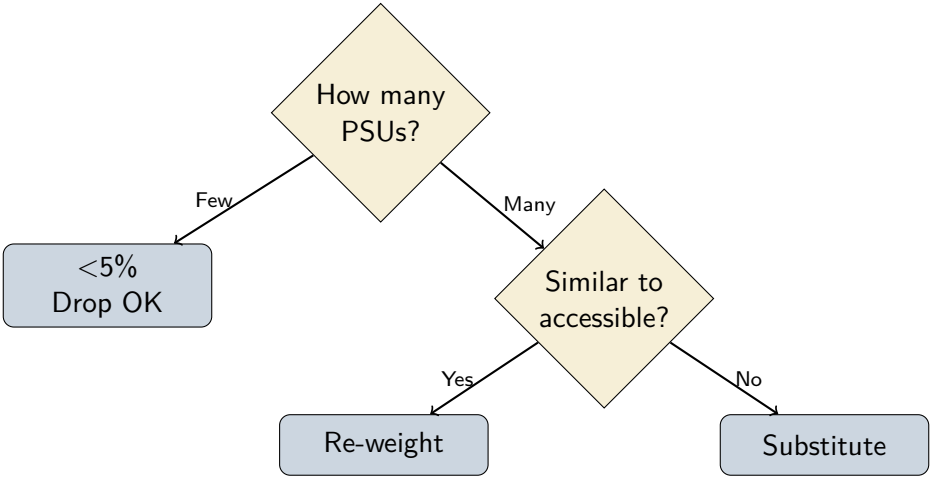
PSUs are inaccessible. What do you do?

Options:

- Drop and proceed
- Re-weight remaining PSUs
- Substitute with similar PSUs
- Use alternative data collection

The right choice depends on **context**.

Adaptive Protocol Decision Tree



Decision 1: How Many PSUs Affected?

Proportion	Impact	Action
$<5\%$	Minimal	Drop acceptable
5–15%	Moderate	Re-weight or substitute
$>15\%$	Severe	Alternative methods needed

Lab 4.3: 50% of one province = 7.3% national $\rightarrow$ Moderate

Decision 2: Are Inaccessible PSUs Similar?

Check auxiliary data:

- Census characteristics
- Administrative records
- Satellite imagery
- Previous survey rounds

If similar: Re-weighting works well.
If different: Re-weighting amplifies bias.

Lab 4.3: Were They Similar?

Characteristic	Accessible	Inaccessible
Mean Income (R)	244,458	291,XXX
% Urban	XX%	XX%
Mean HH Size	3.15	3.21

Inaccessible PSUs had **higher income** — not similar!

This explains why dropping caused **underestimation**.

Decision 3: Is Substitution Feasible?

Requirements for good substitution:

- Available PSUs in same stratum
- Similar characteristics to original
- Logistically accessible
- Time to implement

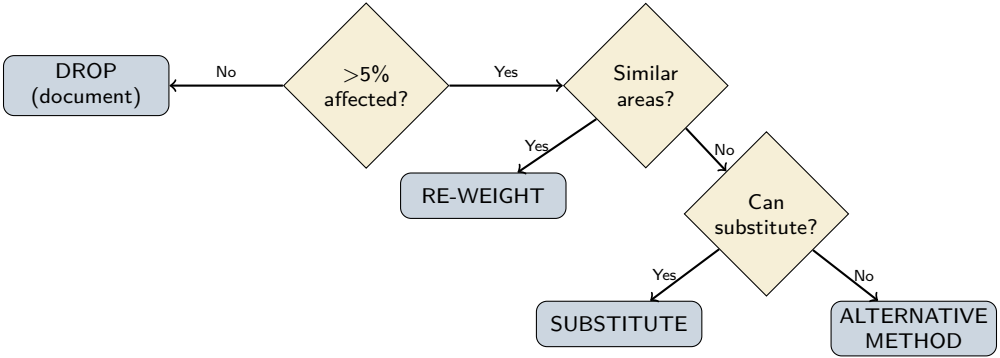
If yes: Substitute with careful matching.
If no: Consider alternative methods.

Decision 4: Alternative Data Collection?

When face-to-face is impossible:

- Phone surveys (CATI)
- SMS/mobile surveys
- Web surveys (urban areas)
- Remote sensing proxies
- Administrative data

Complete Decision Framework



When to Re-weight

Re-weighting is appropriate when:

1. Accessible and inaccessible PSUs are **exchangeable**
2. Inaccessibility is **unrelated** to survey outcomes
3. Sufficient PSUs remain in each **stratum**
4. Design can handle **inflated weights**

Re-weighting: The Mechanics

For each affected stratum:

$$w_{adjusted} = w_{original} \times \frac{\sum_{all} w_i}{\sum_{accessible} w_i}$$

Interpretation: Remaining PSUs "speak for" missing ones.

Lab 4.3 Result: +3.05% bias (better than –6.04% from dropping)

Re-weighting: The Risks

Potential problems:

- **Variance inflation:** Fewer PSUs = larger SE
- **Extreme weights:** Some HH represent many
- **Model dependence:** Assumes exchangeability
- **Hidden bias:** If assumption violated

Always check weight distribution after adjustment.

Lab 4.3: Re-weighting Impact

Metric	Original	Re-weighted
Sample size	19,401	19,401
Weight sum	56.2M	60.8M
Max weight	5,050	10,100
Mean Income (R)	244,458	268,100
Bias	–6.04%	**+3.05%**

Bias reduced by half, but max weight doubled.

When to Substitute

Substitution is appropriate when:

1. Re-weighting would be **inappropriate**
2. Good **matching** PSUs available
3. **Time** permits field reassignment
4. Maintaining **sample size** is critical

Substitution: Finding Good Matches

Matching hierarchy:

1. Same stratum (required)
2. Same geography type
3. Similar PSU size
4. Geographic proximity
5. Similar auxiliary characteristics

Lab 4.3: Random within stratum (simplified).

Substitution: The Risks

Potential problems:

- **Selection bias:** Substitutes may differ
- **Duplication:** Same donor used multiple times
- **Logistics:** Field team redeployment
- **Documentation:** Complex audit trail

Always flag substituted PSUs in the dataset.

Lab 4.3: Substitution Impact

Metric	Dropped	Substituted
Sample size	19,401	20,955
Original PSUs	100%	92.6%
Substitute PSUs	0%	7.4%
Mean Income (R)	244,458	270,020
Bias	–6.04%	**+3.79%**

Sample size maintained, but bias similar to re-weighting.

Alternative Data Collection Methods

When face-to-face is impossible,

what are the alternatives?

Option 1: Phone Surveys (CATI)

Computer-Assisted Telephone Interviewing

Advantages:

- Safe for interviewers
- Fast deployment
- Lower cost per interview

Challenges:

- Phone coverage bias
- Shorter questionnaires
- No visual verification

Phone Surveys: Key Considerations

Coverage: Who has phones?

- Urban > Rural
- Young > Old
- Wealthy > Poor

Mitigation: Weight by phone ownership probability.

Option 2: SMS/Mobile Surveys

Short questionnaires via text message

Advantages:

- Very low cost
- Asynchronous response
- Wide geographic reach

Challenges:

- Very short questionnaires (5–10 items)
- Low response rates
- Literacy requirements

Option 3: Remote Sensing Proxies

Satellite imagery as proxy for survey data

Can measure:

- Roof quality (wealth proxy)
- Agricultural activity
- Night-time lights (economic activity)
- Settlement density

Limitation: Proxy, not direct measurement.

Option 4: Administrative Data

Use existing records as survey substitutes

Sources:

- Tax records
- Utility connections
- Health facility data
- School enrollment

Challenge: Coverage and quality vary.

Choosing Alternative Methods

Method	Cost	Coverage	Data Quality
Face-to-face	High	Best	Best
Phone (CATI)	Medium	Good	Good
SMS/Mobile	Low	Moderate	Limited
Remote sensing	Low	Geographic	Proxy only
Admin data	Very low	Variable	Variable

Trade-off: Cost vs Quality vs Coverage

The Hybrid Approach

Combine methods:

- Face-to-face in accessible areas
- Phone surveys in semi-accessible areas
- Remote sensing for inaccessible areas
- Calibrate to administrative data

Key: Document mode effects and adjust.

Documentation Protocol

For every inaccessible PSU, record:

1. PSU identifier and location
2. Date inaccessibility determined
3. Reason (security, weather, infrastructure)
4. Action taken (drop/reweight/substitute/alternative)
5. Expected bias direction

Sensitivity Analysis Protocol

Always report multiple scenarios:

Scenario	**Estimate (R)**
Best case (re-weight)	268,100
Alternative (substitute)	270,020
Worst case (drop)	244,458
Range	**±12%**

Users understand the uncertainty.

Real-World Examples

Afghanistan: Phone surveys during conflict periods

Syria: Remote sensing for displacement tracking

South Sudan: Adaptive sampling with substitution protocols

DRC: Hybrid face-to-face + phone in unstable areas

Conflict is common — protocols are essential.

Module 4.3: Key Takeaways

1. Use **decision tree** to choose strategy
2. **Re-weight** when PSUs are exchangeable
3. **Substitute** when good matches available
4. Consider **alternative methods** when needed
5. Always do **sensitivity analysis**
6. **Document** everything

Next: Module 4.4 — Quality Assurance

From Crisis Response to Prevention

4.3: Handling problems when they occur

↓

4.4: Building quality into the process

Short Break: 15 minutes

The QA Dashboard

Paradata Forensics and Fraud Detection

Tuesday 2:15 PM: The Coffee Shop

Lindiwe stops at Café Ubuntu for a quick lunch...

She spots a familiar face: **Enumerator ENUM_001**

Sitting alone. Tablet in hand.

Filling out survey forms.

No respondent in sight.

The Realization

Lindiwe watches for 15 minutes...

ENUM_001 completes **four** household questionnaires.

Each takes about **3 minutes**.

The GHS questionnaire should take **35 minutes**.

This is curbstoning.

What is Curbstoning?

Definition: Fabricating survey data without conducting interviews.

Common forms:

- Filling forms in coffee shops, cars, hotels
- Copying data from previous interviews
- Inventing responses entirely
- Interviewing convenient non-sample members

Named for enumerators who sit on the curb instead of knocking on doors.

Why Does Curbstoning Happen?

Motivation:

- Payment per completed interview
- Difficult-to-reach respondents
- Dangerous neighborhoods
- Fatigue and burnout
- Poor supervision

Consequence: Biased estimates, wasted resources, damaged credibility.

The Power of Paradata

Paradata: Data *about* the data collection process.

Automatically captured by CAPI systems:

- Interview start/end timestamps
- Duration per section
- GPS coordinates
- Keystroke patterns
- Number of corrections

Paradata can **reveal** what enumerators try to **hide**.

Lab 4.4: Paradata Forensics

Metric	Value
Total interviews	20,940
Enumerators	50
Flagged (short duration)	526
Flagged (suspicious time)	128
Any flag	**544**

544 interviews (2.6%) triggered at least one flag.

Flag 1: Interview Duration

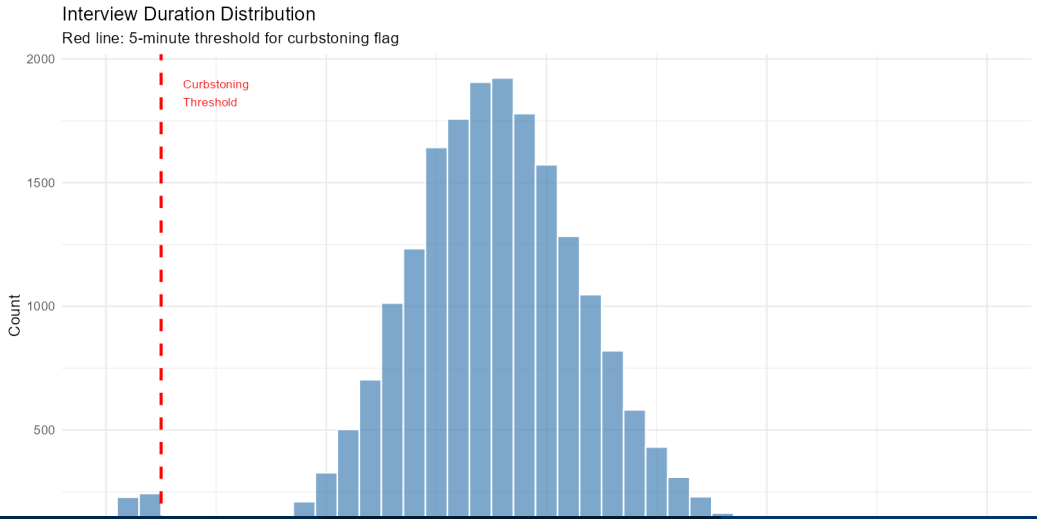
Criterion: Interview < 5 minutes

Rationale: GHS questionnaire cannot be completed properly in under 5 minutes.

Lab 4.4 Result: 526 interviews (2.5%) flagged.

Investigation: Verify with callbacks, check response patterns.

Duration Distribution



The Spike of Suspicion

Notice the cluster below 5 minutes?

Normal interviews: 25–45 minutes (bell curve)

Suspicious interviews: 1–5 minutes (separate cluster)

These are the interviews that need investigation.

526 interviews fall in this zone.

Flag 2: Interview Timestamp

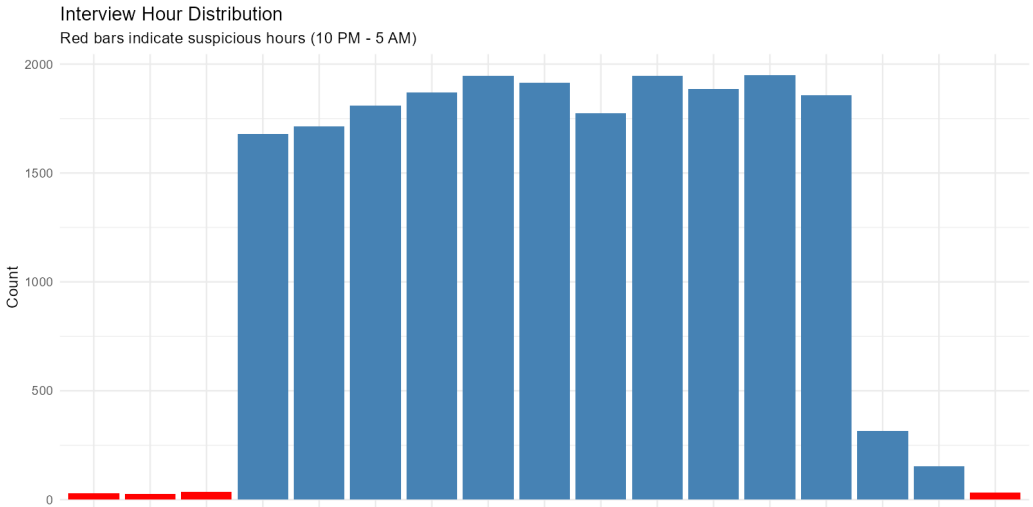
Criterion: Interview conducted 10 PM – 5 AM

Rationale: Household interviews at 2 AM are highly improbable.

Lab 4.4 Result: 128 interviews (0.6%) flagged.

Explanation: Could be timestamp errors, but warrant review.

Interview Hour Distribution



Red Flags at Night

Normal pattern: Interviews clustered 8 AM – 6 PM

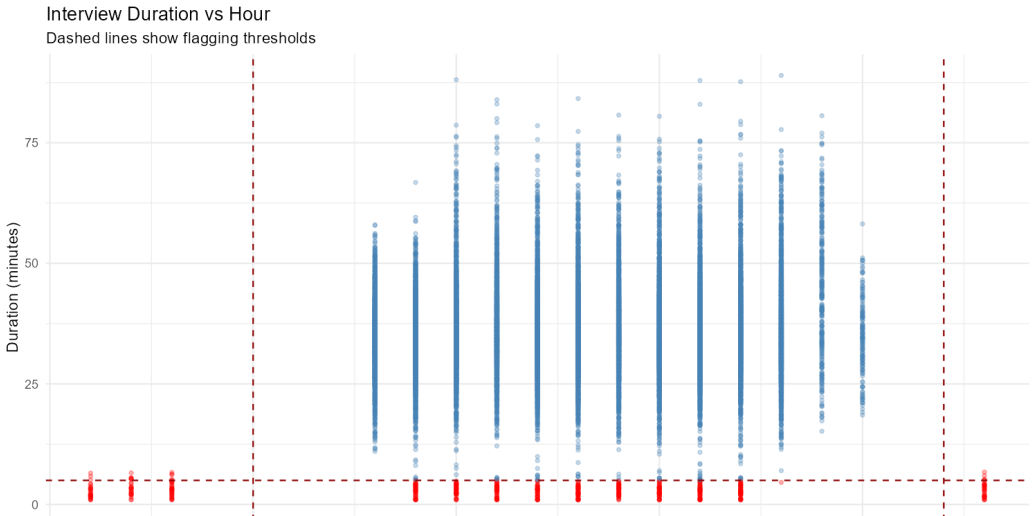
Suspicious: Interviews at 1 AM, 2 AM, 3 AM, 11 PM

Possible explanations:

- Enumerator entering data late at night
- Timestamp manipulation
- System clock errors
- Actual fraud

All require investigation.

Duration vs Hour: The Full Picture



Reading the Scatterplot

The "danger zone":

- Below horizontal line: Duration < 5 min
- Left of vertical line: Before 5 AM
- Right of vertical line: After 10 PM

Red points = Flagged interviews

Worst cases: Short interviews at suspicious hours (bottom corners).

Analyzing by Enumerator

Key question: Are flags concentrated in certain enumerators?

Enumerator	N	Flagged	%
ENUM_001	336	16	4.8%
ENUM_039	396	17	4.3%
ENUM_020	397	15	3.8%
ENUM_023	482	17	3.5%
...	...	...	...

Top enumerators have higher flag rates.

Risk Score Calculation

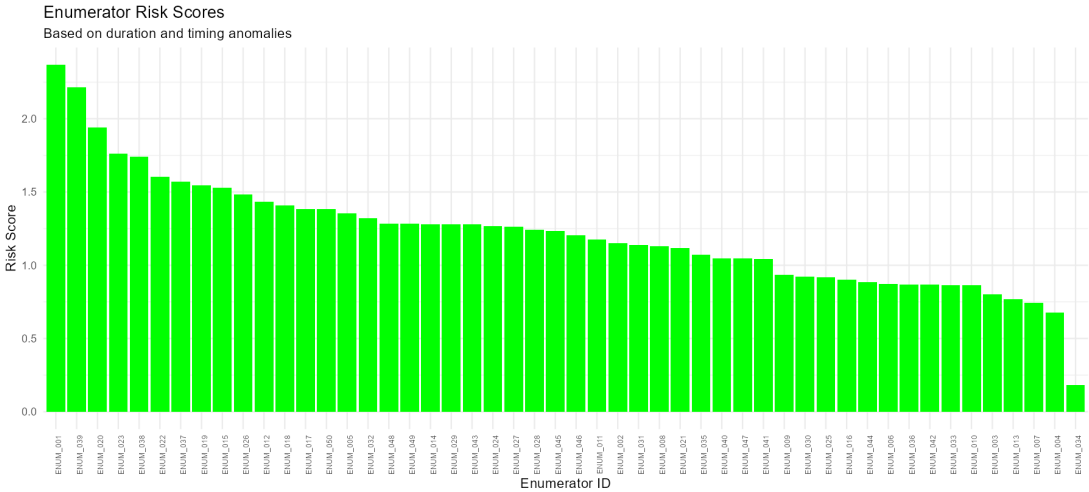
Combining multiple indicators:

$$\text{Risk} = 0.4 \times \%\text{Short} + 0.3 \times \%\text{SuspTime} + 0.2 \times \text{DurPenalty} + 0.1 \times \text{SDPenalty}$$

Categories:

- NORMAL: Score ≤ 5
- LOW: Score 5–15
- MEDIUM: Score 15–30
- HIGH: Score > 30

Enumerator Risk Scores



Back to ENUM_001

The enumerator Lindiwe spotted:

Metric	Value
Total interviews	336
Mean duration	34.7 min
Min duration	1.0 min
Flagged short	16 (4.8%)
Flagged time	3 (0.9%)
Risk score	2.37

Risk score is "NORMAL" — but the coffee shop behavior tells another story.

Limitations of Paradata Alone

Paradata flags are *indicators*, not proof.

False positives:

- Legitimate short interviews (refusals, empty HH)
- System glitches with timestamps
- Battery/connectivity issues

False negatives:

- Clever fabricators time their entries
- Copying from real interviews

Always combine with other verification methods.

Advanced Detection: Benford's Law

Benford's Law: First digits of naturally occurring numbers follow a predictable distribution.

First Digit	Expected %
1	30.1%
2	17.6%
3	12.5%
...	...
9	4.6%

Fabricated data often **fails** this test.

Why Benford's Law Works

Natural data spans orders of magnitude:

Incomes: R 1,000 . . . R 10,000 . . . R 100,000

To go from 1,000 to 2,000 (first digit 2): grow 100%

To go from 9,000 to 10,000 (first digit 1 again): grow 11%

Numbers "spend more time" with first digit 1.

Fabricators: Often distribute digits uniformly.

Applying Benford to Survey Data

Works well for:

- Income values
- Population counts
- Financial transactions
- Distances/areas

Does not work for:

- Age (constrained range)
- Survey codes (assigned numbers)
- Percentages

Other Fraud Detection Methods

GPS verification:

- Was enumerator at assigned location?
- Multiple interviews from same GPS point?

Response pattern analysis:

- Too many "Don't know" responses
- Unusual correlation patterns
- Identical response sequences

Callback verification:

- Re-contact 10% of interviews
- Verify key responses

Building a Real-Time QA Dashboard

Key components:

1. Duration monitoring (flag < 5 min)
2. Timestamp monitoring (flag unusual hours)
3. GPS tracking (flag stationary clusters)
4. Completion rates by enumerator
5. Response pattern alerts

Goal: Catch problems **during** fieldwork, not after.

Sample Dashboard Layout

ZNSO FIELD QUALITY DASHBOARD

Today's Metrics

Interviews: 342
Flagged: 8 (2.3%)
Active Enum: 45

Avg Duration

34.2 min (OK)

Alerts

! ENUM_001: 3 short
! ENUM_039: 2 AM entry
Click for details. . .

GPS Coverage

[Map of PSU locations]

Enumerator Status

ID	Done	Flagged	Risk
001	12	2	2.4
039	15	1	2.2
020	11	0	1.9

Lindiwe Takes Action

After the coffee shop incident:

1. Documents the observation
2. Pulls ENUM_001's paradata
3. Finds 16 interviews under 5 minutes
4. Orders callback verification
5. 12 of 16 cannot be verified
6. ENUM_001 suspended pending investigation

Lesson: Paradata + observation = evidence.

Prevention is Better Than Detection

Reduce curbstoning through:

- Strong training and ethical standards
- Realistic workloads
- Proper payment (not per-interview only)
- Real-time monitoring
- Random field spot-checks
- Callback verification protocols

Culture of quality beats **culture of quotas**.

Module 4.4: Key Takeaways

1. **Paradata** reveals hidden field behavior
2. **Duration flags** catch impossible interviews
3. **Timestamp flags** catch suspicious timing
4. **Risk scores** prioritize investigation
5. **Benford's Law** detects fabricated numbers
6. **Real-time dashboards** enable rapid response

Day 4: Complete

Today we learned to:

- Apply the **Total Survey Error** framework
- Diagnose **non-response bias**
- Implement **hot-deck imputation**
- Manage **conflict zone** surveys
- Build **QA dashboards** with paradata

Tomorrow: Synthesis and Longitudinal Surveys

Systematic Quality Control

Daily Checks and Outlier Detection

The QC Philosophy

Lindiwe implements a new daily routine at ZNSO...

Old approach: Check data quality after fieldwork ends.

New approach: Check data quality **every day** during fieldwork.

Principle: Catch errors when they can still be fixed.

The 5 Daily Checks

1 Completeness

2 Duration

3 Timestamps

4 Outliers

5 Patterns

Every morning, before fieldwork resumes.
Every evening, after data uploads complete.

Check 1: Completeness

DAILY CHECK #1: COMPLETENESS MONITORING

- ☐ Count interviews uploaded vs. target
- ☐ Calculate completion rate by PSU
- ☐ Identify PSUs with zero uploads
- ☐ Flag enumerators below daily target
- ☐ Document reasons for shortfalls

Threshold: Alert if completion $< 80\%$ of daily target.

Completeness Metrics

Date	Target	Actual	%
March 1	700	685	97.9%
March 2	700	712	101.7%
March 3	700	543	77.6%
March 4	700	698	99.7%

March 3: Weather event in Northern Province. Documented.

Check 2: Duration Monitoring

DAILY CHECK #2: DURATION MONITORING

- ☐ Calculate mean/median duration
- ☐ Count interviews < 5 minutes
- ☐ Count interviews > 90 minutes
- ☐ Identify enumerators with abnormal means
- ☐ Generate duration histogram

Thresholds:

- Flag if < 5 min (too short)
- Flag if > 90 min (data entry issues?)

Duration: Standard Operating Procedure

If interview $<$ 5 minutes:

1. Check if refusal or vacant household
2. If neither, flag for callback verification
3. Notify field supervisor

If interview $>$ 90 minutes:

1. Check for data entry interruptions
2. Verify questionnaire completeness
3. Review with enumerator

Check 3: Timestamp Validation

DAILY CHECK #3: TIMESTAMP VALIDATION

- ☐ Count interviews outside 7 AM – 9 PM
- ☐ Check for future dates
- ☐ Check for duplicate timestamps
- ☐ Verify GPS timestamp alignment
- ☐ Flag interviews at impossible times

Threshold: Any interview 10 PM – 5 AM requires explanation.

Timestamp Red Flags

Immediate investigation required:

- Interview at 2:00 AM
- 10 interviews with identical timestamps
- Timestamps in the future
- GPS timestamp 3+ hours from interview timestamp

Lab 4.4 found: 128 interviews at suspicious hours (0.6%).

Check 4: Outlier Detection

DAILY CHECK #4: OUTLIER DETECTION

- ☐ Calculate IQR for key numeric variables
- ☐ Flag values $> Q3 + 3\times IQR$
- ☐ Flag values $< Q1 - 3\times IQR$
- ☐ Review outliers for plausibility
- ☐ Document legitimate extreme values

Focus variables: Income, household size, expenditure.

Check 5: Pattern Analysis

DAILY CHECK #5: PATTERN ANALYSIS

- ☐ Check skip pattern consistency
- ☐ Verify logical relationships
- ☐ Compare distributions to expectations
- ☐ Flag identical response sequences
- ☐ Review item non-response rates

Example: Age = 5, Employment = "Full-time employed" → Error!

The Daily QC Report

Every morning, the QC team produces:

Section	Content
1. Summary	Yesterday's key metrics
2. Completeness	Progress vs. target
3. Duration flags	List of short/long interviews
4. Timestamp flags	Suspicious timing
5. Outlier list	Extreme values for review
6. Action items	Required follow-ups

Outlier Detection: Deep Dive

Focus: Income Variables

Income data is particularly prone to:

- Reporting errors (extra zeros)
- Unit confusion (monthly vs. annual)
- Refusal-driven missing data
- Legitimate extreme values

Method 1: IQR-Based Outliers

Interquartile Range Method:

$$\text{Lower fence} = Q_1 - 1.5 \times IQR$$

$$\text{Upper fence} = Q_3 + 1.5 \times IQR$$

Extreme outliers:

$$\text{Lower extreme} = Q_1 - 3 \times IQR$$

$$\text{Upper extreme} = Q_3 + 3 \times IQR$$

IQR Example: Household Income

GHS 2024 Income Distribution:

Statistic	**Value (R)**
Q1 (25th percentile)	1,500
Median (Q2)	5,257
Q3 (75th percentile)	15,000
IQR	13,500
Upper fence (1.5×IQR)	35,250
Upper extreme (3×IQR)	55,500

Income $>$ R 55,500 $=$ extreme outlier for review.

Method 2: Z-Score Outliers

Standard deviation method:

$$z = \frac{x - \bar{x}}{s}$$

Thresholds:

- $|z| > 2$: Potential outlier (5% expected)
- $|z| > 3$: Probable outlier (0.3% expected)
- $|z| > 4$: Extreme outlier (rare)

Limitation: Sensitive to skewed distributions.

Method 3: MAD-Based Outliers

Median Absolute Deviation (robust):

$$MAD = \text{median}(|x_i - \text{median}(x)|)$$

$$\text{Modified } z = \frac{0.6745 \times (x_i - \text{median})}{MAD}$$

Advantage: Resistant to extreme values.

Threshold: Modified $z > 3.5 =$ outlier.

Choosing an Outlier Method

Method	Best For	Limitation
IQR	Skewed data	Arbitrary thresholds
Z-score	Normal data	Affected by outliers
MAD	Highly skewed	Less familiar

Recommendation: Use IQR for income (skewed distribution).

Always **review** flagged values — don't auto-delete!

R Code: Outlier Detection

```
# IQR method
Q1 <- quantile(ghs$totmhinc, 0.25, na.rm = TRUE)
Q3 <- quantile(ghs$totmhinc, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1

lower_fence <- Q1 - 1.5 * IQR
upper_fence <- Q3 + 1.5 * IQR

ghs$outlier_income <- ghs$totmhinc < lower_fence |
                      ghs$totmhinc > upper_fence

# Count outliers
sum(ghs$outlier_income, na.rm = TRUE)
```

STATA Code: Outlier Detection

```
* IQR method
summarize totmhinc, detail
local Q1 = r(p25)
local Q3 = r(p75)
local IQR = `Q3' - `Q1'

local lower = `Q1' - 1.5 * `IQR'
local upper = `Q3' + 1.5 * `IQR'

gen outlier_income = (totmhinc < `lower') | ///
                     (totmhinc > `upper')

* Count outliers
count if outlier_income == 1
```

Outlier Review Process

For each flagged outlier:

1. Is the value **plausible**? (CEO income vs. typo)
2. Is it **consistent** with other variables?
3. Can it be **verified** by callback?
4. Should it be **corrected**, **kept**, or **flagged**?

Never delete outliers without documentation.

Common Income Data Errors

Error types to look for:

Error	Example	Fix
Extra zero	R 50,000 $\rightarrow$ R 500,000	Divide by 10
Missing zero	R 5,000 $\rightarrow$ R 500	Multiply by 10
Annual/monthly	R 120,000/yr as monthly	Divide by 12
Currency confusion	USD entered as ZAR	Convert

Key: Check auxiliary variables (occupation, HH size).

Plausibility Checks

Cross-validate income with:

- **Occupation:** Farm laborer with R 500,000/month?
- **Education:** No schooling with R 200,000/month?
- **Housing:** Informal dwelling with R 100,000/month?
- **Assets:** No car, no appliances, high income?

Inconsistency $\neq$ error, but warrants review.

Daily QC Checklist: Complete

ZNSO DAILY QUALITY CONTROL CHECKLIST

Date: _____ QC Officer: _____

CHECK 1: COMPLETENESS

☐ Interviews uploaded: _____ / _____ target (_____%)

☐ PSUs with zero uploads: _____ (list attached)

CHECK 2: DURATION

☐ Mean duration: _____ min Median: _____ min

☐ Interviews $<$ 5 min: _____ $>$ 90 min: _____

CHECK 3: TIMESTAMPS

☐ Interviews 10PM–5AM: _____

☐ Future dates: _____ Duplicates: _____

CHECK 4: OUTLIERS

☐ Income outliers: _____ (list attached)

☐ HH size outliers: _____

CHECK 5: PATTERNS

☐ Skip errors: _____ Logic errors: _____

SIGNATURE: _____

Escalation Protocol

Level 1 — Field Supervisor:

- 1–5 flagged interviews per enumerator
- Single outlier requiring verification

Level 2 — Provincial Coordinator:

- >5 flags per enumerator
- Pattern of issues in one PSU

Level 3 — Survey Manager (Lindiwe):

- Suspected systematic fraud
- $>10\%$ of enumerator's work flagged
- Provincial completion $< 70\%$

Documentation Standards

Every QC action must record:

1. **What** was flagged (variable, value, ID)
2. **Why** it was flagged (rule triggered)
3. **Who** reviewed it
4. **Decision** made (keep/correct/delete/impute)
5. **Justification** for decision

Audit trail: Essential for reproducibility.

Building QC into CAPI Systems

Real-time validation:

- Range checks (income 0–10,000,000)
- Soft warnings (income $>$ 100,000)
- Hard stops (age $<$ 0)
- Cross-variable checks

Advantage: Errors caught at source.

Risk: Over-restriction leads to false data entry.

The QC Mindset

Quality Control is not:

- Punishment for enumerators
- Finding someone to blame
- Adding bureaucratic burden

Quality Control is:

- Protecting data integrity
- Supporting field staff
- Ensuring policy relevance
- Maintaining public trust

Systematic QC: Key Takeaways

1. Run **5 daily checks** every day during fieldwork
2. Use **IQR method** for skewed income data
3. **Review** outliers — don't auto-delete
4. **Document** every QC decision
5. **Escalate** systematic issues immediately
6. Build checks into **CAPI systems**

Day 4: Module 4.4 Complete

Quality Assurance Integration

Today's QA toolkit:

- Paradata forensics (duration, timestamps)
- Curbstoning detection
- Benford's Law
- 5 daily QC checks
- Outlier detection methods
- Documentation standards

End of Day 4 — See you tomorrow for Synthesis!

Practical Exercise Review

Labs 4.2, 4.3, and 4.4 — What Did We Learn?

Review Objectives

In this session, we will:

1. Walk through Lab 4.2 results (Hot-Deck Imputation)
2. Walk through Lab 4.3 results (Conflict Zone Strategies)
3. Discuss Lab 4.4 findings (Paradata Forensics)
4. Explore **ethical implications** of data cleaning decisions

This is interactive — be ready to discuss!

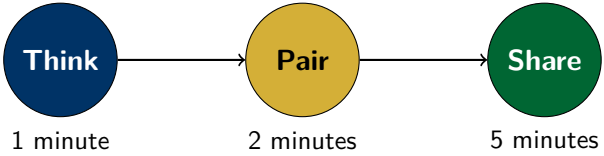
The Big Picture

Today's exercises showed that:

- Data cleaning is **not neutral** — every choice has consequences
- Different methods produce **different answers**
- The "right" method depends on **context and assumptions**
- **Transparency** is essential

Key insight: There is no "clean" data — only data with documented decisions.

Discussion Format



For each question: Think alone → Discuss with neighbor → Share with group

Lab 4.2: Hot-Deck Imputation Recap

The scenario:

- Income variable with 26.4% missing (5,529 of 20,940)
- Compared three approaches:
  1. Complete cases only (listwise deletion)
  2. Mean imputation
  3. Hot-deck imputation

Question: Which method gave the best results?

Lab 4.2: The Results

Method	N	Mean (R)	SD (R)	Bias
True values	20,940	218,305	1,421,646	—
Complete cases	15,411	214,421	1,409,264	-1.78%
Mean imputation	20,940	214,421	1,208,970	-1.78%
Hot-deck	**20,940**	**223,244**	**1,439,248**	$+2.26\%$

Discussion: Which method would you choose and why?

Lab 4.2: The Variance Problem

Method	SD (R)	SD Change
True values	1,421,646	—
Complete cases	1,409,264	-0.9%
Mean imputation	1,208,970	-15.0%
Hot-deck	1,439,248	$+1.2\%$

Key finding: Mean imputation **destroyed 15%** of the variance!

Why does this matter for policy analysis?

Think-Pair-Share #1

DISCUSSION QUESTION

Mean imputation gave the **same mean** as complete cases, but **destroyed variance**.

Hot-deck preserved variance but introduced **+2.26% bias**.

Which trade-off is more acceptable for poverty analysis?

Think: 1 min → Pair: 2 min → Share: 5 min

Lab 4.2: The Median Disaster

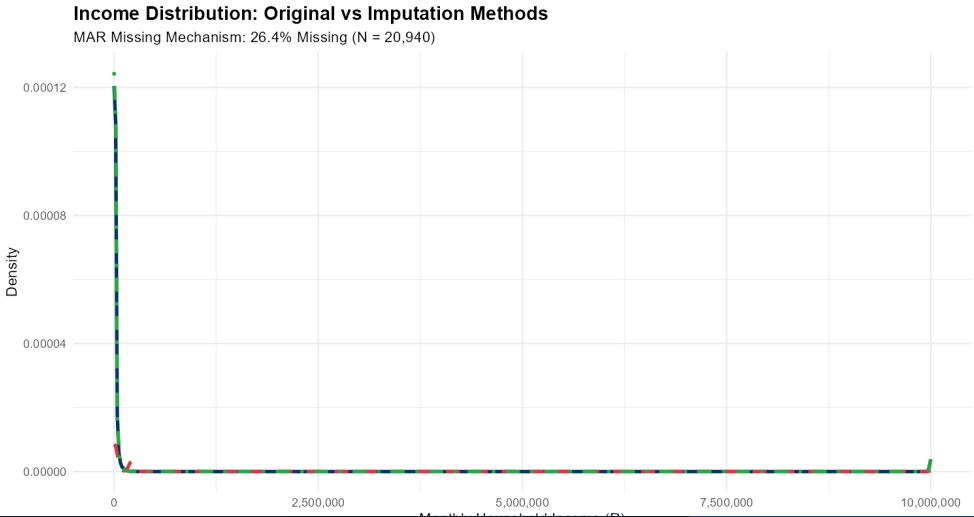
Method	Median (R)	Change
True values	5,257	—
Complete cases	5,180	-1.5%
Mean imputation	8,960	$+70.4\%$
Hot-deck	5,180	-1.5%

Mean imputation inflated the median by 70%!

5,529 households received identical value of R 214,421.

This shifted the entire distribution.

Lab 4.2: Visual Evidence



Lab 4.2: Why Hot-Deck Worked

Hot-deck preserved the distribution because:

1. Values came from **real donors** (actual survey responses)
2. Donors matched on **Province** $\times$ **HH Size** (36 classes)
3. Random selection preserved **natural variability**
4. No artificial clustering at a single value

Trade-off: Small mean bias (+2.26%) vs. preserved distribution.

Lab 4.2: Key Lessons

1. **Mean imputation** is easy but dangerous
2. **Variance matters** — not just the mean
3. **Distribution shape** affects policy conclusions
4. **Hot-deck** is a reasonable single-imputation method
5. **Multiple imputation** is even better (but more complex)

Bottom line: Never use mean imputation for skewed distributions.

Lab 4.3: Conflict Zone Strategies Recap

The scenario:

- Eastern Highlands (KwaZulu-Natal) conflict zone
- 50% of PSUs inaccessible
- 1,539 households lost (7.3% of national sample)
- Compared three response strategies

Question: How should we handle inaccessible areas?

Lab 4.3: The Results

Strategy	N	Mean Income (R)	Bias
True (full sample)	20,940	260,166	—
A: Drop inaccessible	19,401	244,458	-6.04%
B: Re-weight	19,401	268,100	$+3.05\%$
C: Substitute	20,955	270,020	$+3.79\%$

Discussion: Which strategy would you recommend?

Lab 4.3: Why Dropping Failed

Strategy A (Drop) had the worst bias: -6.04%

Why?

- Conflict zone was **not random** — it was in wealthier urban areas
- Dropping systematically excluded **higher-income households**
- Remaining sample was **no longer representative**

Lesson: "Doing nothing" is still a choice — and often the worst one.

Lab 4.3: Re-weighting Trade-off

Strategy B (Re-weight) had +3.05% bias

How it works:

- Increase weights of accessible PSUs in same stratum
- Assume accessible $\approx$ inaccessible (within stratum)

Risk: If inaccessible areas are systematically different, re-weighting amplifies the bias.

In this case: Safe zone was poorer than conflict zone, so re-weighting *overestimated* income.

Lab 4.3: Substitution Trade-off

Strategy C (Substitute) had +3.79% bias

How it works:

- Replace inaccessible PSUs with "similar" accessible ones
- Maintains sample size
- Doubles representation of some areas

Risk: Substitutes may not be truly similar.

In this case: Similar bias to re-weighting, but maintained sample size.

Think-Pair-Share #2

DISCUSSION QUESTION

Strategy B (re-weight): +3.05% bias, N = 19,401
Strategy C (substitute): +3.79% bias, N = 20,955

Is it worth +0.74% extra bias to maintain sample size?

What other factors should influence this decision?

Think: 1 min → Pair: 2 min → Share: 5 min

Lab 4.3: Key Lessons

1. **Dropping** is rarely the best option
2. **Re-weighting** assumes similarity within strata
3. **Substitution** maintains size but may duplicate bias
4. **All methods** require assumptions — document them!
5. **Sensitivity analysis** is essential

Bottom line: There's no perfect solution — choose the least bad option and document why.

Lab 4.4: Paradata Forensics Recap

The scenario:

- Analyzed 20,940 interviews from 50 enumerators
- Flagged suspicious patterns in duration and timing
- Generated risk scores for each enumerator

Flag Type	**Count (%)**
Short duration (<5 min)	526 (2.5%)
Suspicious time (10PM–5AM)	128 (0.6%)
Any flag	544 (2.6%)

Lab 4.4: Key Lessons

1. **Paradata** reveals hidden field behavior
2. **Flags** are indicators, not proof
3. **Enumerator patterns** matter more than individual interviews
4. **Real-time monitoring** catches problems early
5. **Documentation** protects everyone

Bottom line: Quality assurance is prevention, not punishment.

From Methods to Ethics

We've discussed **how** to clean data.

Now let's discuss **whether** we should.

Every data cleaning decision has ethical implications.

The Ethics of Data Cleaning

Data cleaning is not just technical — it's ethical.

Key ethical questions:

- Who benefits from our cleaning decisions?
- Who might be harmed?
- Are we being transparent about our choices?
- Would we be comfortable if our decisions were public?

Ethical Dilemma #1: Imputation

SCENARIO:

You impute income for 5,529 households.
Your imputed values will determine their eligibility for a social grant.

Hot-deck overestimates income by 2.26%.
Some households may be incorrectly classified as "too wealthy" for assistance.

Is it ethical to use imputed values for benefit eligibility?

Think-Pair-Share #3

DISCUSSION QUESTION

If imputation affects grant eligibility:

Option A: Use imputed values anyway (consistent methodology)
Option B: Exclude imputed cases from eligibility analysis
Option C: Flag imputed cases for manual review

Which approach is most ethical? Why?

Think: 1 min → Pair: 2 min → Share: 5 min

Ethical Dilemma #2: Conflict Zones

SCENARIO:

Your survey couldn't reach conflict-affected areas.
These areas may have the **greatest need** for services.

If you drop them: They become **invisible** to policy.
If you substitute: You may **misrepresent** their situation.

Is it better to have biased data or no data?

Ethical Dilemma #3: Curbstoning

SCENARIO:

You identify an enumerator with suspicious patterns.
They are a single parent supporting three children.
Firing them will cause significant hardship.

But their data may be fabricated.

What is the ethical response?

Principles for Ethical Data Cleaning

1. **Transparency**
   - Document every decision
   - Explain assumptions clearly
2. **Proportionality**
   - Match intervention to problem severity
   - Don't over-correct
3. **Accountability**
   - Be prepared to defend your choices
   - Consider affected populations

Day 4: Complete Summary

Today we learned:

- **Total Survey Error** framework for diagnosing problems
- **Non-response** analysis and bias detection
- **Hot-deck imputation** for item non-response
- **Adaptive strategies** for conflict zones
- **Paradata forensics** for quality assurance
- **Ethical dimensions** of data cleaning

Core message: Every cleaning decision shapes the story your data tells.

Looking Ahead: Day 5

Day 5: Synthesis and Longitudinal Surveys

Topics:

- Panel survey design
- Rotation patterns
- Attrition management
- Longitudinal weights
- Documentation standards
- Implementation roadmap

See you tomorrow at 09:00!

Day 4: Conclusion

Error Mitigation and Quality Assurance

Lindiwe's Journey Today

What Lindiwe learned on Day 4...

- Morning: Diagnosed survey errors using the TSE framework
- Mid-morning: Discovered 26.4% income data was missing
- Afternoon: Implemented hot-deck imputation
- Late afternoon: Managed a conflict zone crisis
- Evening: Caught an enumerator fabricating data

A full day of error mitigation!

Day 4 Learning Objectives: Achieved?

Learning Objective	Status
LO4.1: Apply TSE framework to diagnose errors	✓
LO4.2: Implement non-response weight adjustments	✓
LO4.3: Apply hot-deck imputation	✓
LO4.4: Adapt methodology for fragile settings	✓
LO4.5: Build QA systems with paradata	✓

All objectives covered!

Recap: Module 4.1 — Total Survey Error

Key concepts:

- TSE = Sampling Error + Non-Sampling Error
- Non-sampling errors often **larger** than sampling errors
- Four error sources: Coverage, Sampling, Non-response, Measurement
- Diagnosis before treatment

Zambara application: Lindiwe used TSE to systematically identify error sources in the GHS before attempting corrections.

Recap: Module 4.2 — Non-Response and Imputation

Key concepts:

- Unit vs. item non-response
- MCAR, MAR, MNAR mechanisms
- Mean imputation: Easy but **destroys variance**
- Hot-deck imputation: Preserves distribution
- Imputation classes: Province $\times$ HH Size $=$ 36 classes

Lab 4.2 result: Hot-deck preserved SD ($+1.2\%$ change) vs. mean imputation (-15% change).

Recap: Module 4.3 — Adaptive Design

Key concepts:

- Conflict zones require adaptive protocols
- Three strategies: Drop, Re-weight, Substitute
- Dropping is usually the **worst** option
- Decision trees guide protocol selection
- Documentation is essential

Lab 4.3 result: Dropping caused -6.04% bias; re-weighting limited bias to $+3.05\%$.

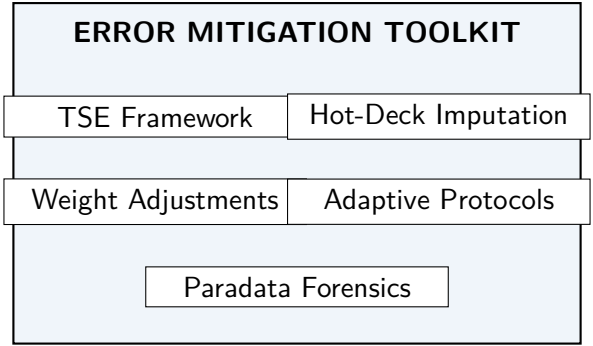
Recap: Module 4.4 — Quality Assurance

Key concepts:

- Paradata reveals hidden field behavior
- Five daily QC checks: Completeness, Duration, Timestamps, Outliers, Patterns
- Risk scores identify problematic enumerators
- Benford's Law detects fabricated numbers
- Real-time dashboards enable rapid response

Lab 4.4 result: 544 interviews (2.6%) flagged for review.

The Error Mitigation Toolkit



Day 4: Key Statistics Summary

Lab	Finding	Value
4.2	Missing income rate	26.4%
4.2	Mean imputation SD change	-15.0%
4.2	Hot-deck SD change	$+1.2\%$
4.2	Hot-deck mean bias	$+2.26\%$
4.3	Households lost to conflict	1,539 (7.3%)
4.3	Drop strategy bias	-6.04%
4.3	Re-weight strategy bias	$+3.05\%$
4.4	Interviews flagged	544 (2.6%)
4.4	Short duration flags	526 (2.5%)

Day 4: The Big Lessons

1. **Errors are inevitable** — mitigation is essential
2. **Mean imputation is dangerous** — use hot-deck instead
3. **Dropping data is rarely optimal** — adjust weights or substitute
4. **Paradata catches fraud** — monitor in real-time
5. **Document everything** — transparency protects credibility
6. **Ethics matter** — cleaning decisions affect real people

Tonight's Homework

The Imputation Challenge

Homework: The Imputation Challenge

Your mission:

You will receive a dataset with **40% missing data** on a key variable.

Your task:

1. Diagnose the missingness pattern (MCAR/MAR/MNAR)
2. Choose an imputation method
3. Implement and evaluate your solution
4. Document your reasoning

Time: 2–3 hours tonight

The Dataset: Zambara Education Survey

File: `zambara_education_hw.dta`

Variable	Description
`hh_id`	Household identifier
`province`	Province (1–9)
`urban`	Urban/Rural indicator
`hh_size`	Household size
`edu_expend`	**Education expenditure (40% missing)**
`income`	Household income
`n_children`	Number of school-age children

N = 5,000 households

Task 1: Diagnose the Missingness

Questions to answer:

1. What is the overall missing rate?
2. Does missingness vary by province? Urban/rural?
3. Is missingness related to income? Household size?
4. What mechanism best explains the pattern?

Deliverable: A table showing missing rates by key subgroups.

Hint: Use `tab province missing` in STATA or `table()` in R.

Task 2: Choose Your Method

Options to consider:

Method	When to Use
Complete cases	MCAR only, low missing rate
Mean imputation	Never (just kidding... almost never)
Hot-deck	MAR, categorical auxiliary vars
Regression imputation	MAR, continuous predictors
Multiple imputation	Best practice, any MAR

Deliverable: Justify your method choice in 2–3 sentences.

Task 3: Implement Your Solution

If you choose hot-deck:

- Define imputation classes (Province $\times$ Urban $\times$ HH Size?)
- Ensure sufficient donors per class
- Implement random donor selection

If you choose regression:

- Build prediction model with auxiliary variables
- Add residual variance (stochastic imputation)

Deliverable: Working R or STATA code with comments.

Task 4: Evaluate Your Results

Compare before and after:

Metric	Complete Cases	After Imputation
N	?	5,000
Mean	?	?
Median	?	?
SD	?	?

Key question: Did your method preserve the distribution shape?

Deliverable: Summary table + density plot comparison.

Homework Submission Requirements

Submit by 08:30 tomorrow:

1. **Code file** (.R or .do) with comments
2. **Brief report** (1–2 pages) containing:
   - Missingness diagnosis (table)
   - Method justification (2–3 sentences)
   - Results comparison (table + plot)
   - Limitations acknowledged
3. **Output files:** Imputed dataset + density plot

Format: Email to workshop coordinator or submit via USB.

Evaluation Criteria

Criterion	**Points**
Correct diagnosis of missingness	20
Appropriate method selection	20
Working implementation	25
Proper evaluation	20
Clear documentation	15
Total	**100**

Bonus points: Compare multiple methods (+10).

Hints and Tips

Common pitfalls to avoid:

- Don't use mean imputation (we know you're tempted!)
- Check for empty imputation classes
- Don't forget to set a random seed
- Compare distributions, not just means
- Document your assumptions

Resources: Lab 4.2 scripts are your template!

Preview: Day 5

Synthesis and Longitudinal Surveys

The Future of Survey Methodology

Why Longitudinal Surveys?

Cross-sectional surveys tell us: "What is happening now?"

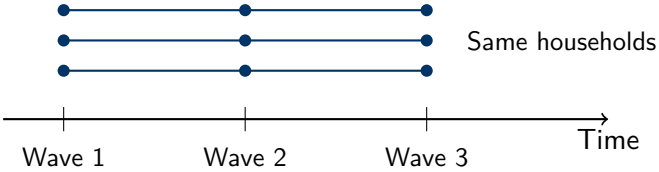
Longitudinal surveys tell us:

- How do things **change** over time?
- Who **enters** and **exits** poverty?
- What are the **pathways** to employment?
- Do interventions have **lasting effects**?

The shift: From snapshots to movies.

Panel Survey Basics

Definition: Following the **same units** over **multiple time points**.



The Attrition Challenge

Problem: People drop out of panel surveys.

Reasons for attrition:

- Migration (can't be found)
- Refusal (don't want to participate)
- Death
- Household dissolution

Consequence: Remaining sample may be **biased**.

Day 5 solution: Attrition weights and tracking protocols.

Rotation Patterns

Not all panels follow everyone forever.

Rotation designs:

- **2-2-2:** In 2 months, out 2 months, in 2 months
- **4-8-4:** In 4 months, out 8 months, in 4 months
- **Continuous:** Replace portion each wave

Trade-off: Respondent burden vs. longitudinal power.

Longitudinal Weights

Challenge: Weights must account for:

1. Original selection probability
2. Non-response at each wave
3. Attrition between waves
4. Population changes over time

Day 5 content:

- Cross-sectional vs. longitudinal weights
- Wave-specific adjustments
- Calibration to population totals

Documentation Standards

Day 5 will also cover:

- Technical documentation requirements
- Metadata standards
- Data dictionaries
- Methodological reports
- Archiving for reproducibility

Goal: Your survey should be reproducible 20 years from now.

Day 5 Schedule Preview

Time	Topic
09:00–10:30	Panel Survey Design
10:30–11:00	*Tea Break*
11:00–12:30	Attrition and Longitudinal Weights
12:30–14:00	*Lunch*
14:00–15:30	Documentation Standards
15:30–16:00	*Tea Break*
16:00–17:00	Synthesis and Implementation Roadmap
17:00–17:30	Closing Ceremony

Lindiwe's Final Challenge

Tomorrow, Lindiwe faces her biggest test...

The Minister wants **annual tracking** of household welfare.

Lindiwe must design:

- A rotation pattern that balances burden and power
- Tracking protocols for mobile populations
- Longitudinal weights that adjust for attrition
- Documentation that will outlast her tenure

Can she build a survey system for the future?

Day 4: Complete

Thank you for your engagement today!

Tonight:

- Complete the Imputation Challenge homework
- Review today's materials
- Rest well!

Tomorrow: Day 5 begins at **09:00 sharp**

See you tomorrow!